COMPARATIVE ANALYSIS OF MECHANICAL PROPERTIES OF HYBRID COMPOSITE REINFORCED WITH GLASS FIBER AND PADDY STRAW IN EPOXY MATRIX FOR STRUCTURAL APPLICATIONS

Design Capstone Project - II

Submitted in Fulfillment of the

Requirement for Award of the Degree

Of

BACHELOR OF TECHNOLOGY

In

AUTOMOBILE ENGINEERING

By

ALIF MD KHAIRUL ARSHI

(12110776)

Under the Guidance of

DR. AKASH GUPTA
(UID-19350)



DEPARTMENT OF MECHANICAL ENGINEERING

LOVELY PROFESSIONAL UNIVERSITY

Lovely Professional University, Jalandhar, Punjab

CERTIFICATE

I hereby certify that the work which is being presented in the capstone entitled **"Comparative Analysis of Mechanical Properties of Hybrid Composite Reinforced with Glass Fiber and Paddy Straw in Epoxy Matrix for Structural Applications"** in partial fulfillment of the requirement for the award of degree of **Bachelor of Technology in Automobile Engineering** and submitted in Department of Mechanical Engineering, Lovely Professional University, Punjab is an authentic record of my own work carried out during period of Capstone under the supervision of **Dr. Akash Gupta**, **Assistant Professor**, Department of Mechanical Engineering, Lovely Professional University, Punjab.

The matter presented in this capstone has not been submitted by me anywhere for the award of any other degree or to any other institute.

Date: **Alif MD Khairul Arshi**

This is to certify that the above statement made by the candidate is correct to best of my knowledge.

Date: **Dr. Akash Gupta**

Supervisor

The B.Tech capstone examination of Alif MD Khairul Arshi, has been held on _____

Signature of Examiner

ACKNOWLEDGEMENT

I would like to extend my heartfelt gratitude to my esteemed professors and mentors at Lovely Professional University for their invaluable support and encouragement throughout this project. Their guidance and insights have been instrumental in shaping the direction of this study.

I am also deeply thankful to my family, peers, and friends for their unwavering encouragement, patience, and belief in my abilities. This work is a testament to the collective effort and collaboration of everyone involved.

Alif MD Khairul Arshi (12110776)

TOPIC APPROVAL PERFORMA

School of Mechanical Engineering (SME)

Program : P139::B.Tech. (Automobile Engineering)

COURSE CODE : MEC582 **REGULAR/BACKLOG :** Regular **GROUP NUMBER :** MERGC0010

Supervisor Name : Dr. Akash Gupta **UID :** 19350 **Designation :** Assistant Professor

Qualification : _____ **Research Experience :** _____

SR.NO.	NAME OF STUDENT	Prov. Regd. No.	BATCH	SECTION	CONTACT NUMBER
1	Alif Md. Khairul Arshi	12110776	2021	M2191	8360331967

SPECIALIZATION AREA : Design **Supervisor Signature:** _____

PROPOSED TOPIC : Comparative Analysis of Mechanical Properties of Hybrid Composite Reinforced with Glass Fiber and Paddy Straw in Epoxy Matrix for Structural Applications

Qualitative Assessment of Proposed Topic by PAC		
Sr.No.	**Parameter**	**Rating (out of 10)**
1	Project Novelty: Potential of the project to create new knowledge	7.77
2	Project Feasibility: Project can be timely carried out in-house with low-cost and available resources in the University by the students.	8.32
3	Project Academic Inputs: Project topic is relevant and makes extensive use of academic inputs in UG program and serves as a culminating effort for core study area of the degree program.	8.55
4	Project Supervision: Project supervisor's is technically competent to guide students, resolve any issues, and impart necessary skills.	8.32
5	Social Applicability: Project work intends to solve a practical problem.	8.32
6	Future Scope: Project has potential to become basis of future research work, publication or patent.	8.55

PAC Committee Members		
PAC Member (HOD/Chairperson) Name: Dr. Jaiinder Preet Singh	UID: 14740	Recommended (Y/N): Yes
PAC Member (Allied) Name: Dr. Rameshwar Cambow	UID: 20345	Recommended (Y/N): Yes
PAC Member 3 Name: Dr. Akash Gupta	UID: 19350	Recommended (Y/N): Yes

Final Topic Approved by PAC: **Comparative Analysis of Mechanical Properties of Hybrid Composite Reinforced with Glass Fiber and Paddy Straw in Epoxy Matrix for Structural Applications**

Overall Remarks: Approved

PAC CHAIRPERSON Name: 11108::Dr. Ankur Bahl **Approval Date:** 20 Nov 2024

5/1/2025 4:11:13 PM

ABSTRACT

The growing requirement of sustainable materials for contemporary engineering, especially for transport applications, has necessitated investigation into the possibilities of environment-friendly composite alternatives. Conventional materials like polycarbonate and ABS, utilized for two-wheeler visors, raise environmental concerns based on their non-biodegradability, limited resistance to UV degradation, and narrow recyclability opportunities. This study explores the production of a fiber-reinforced polymer (FRP) composite with paddy straw agro-waste and glass fiber having an epoxy resin matrix, supplemented further with nano-silica fillers to overcome these shortcomings and enhance mechanical properties.

The study aimed to compare the effect of varying stacking sequences and nano-silica reinforcement on the performance of the composite. Through hand layup process and compression molding, various configurations of composites were prepared such as symmetrical and hybrid glass-paddy layering. Basic mechanical tests—tensile, flexural, and water absorption—were carried out to measure the performance of each configuration. The study outcome showed that the S4 configuration (G/G/G/G) attained maximum tensile strength, with a 198% increase compared to the hybrid composites modified, thanks to its uniform strain distribution and higher load-carrying capability. The S3 configuration (G/P/P/G) also had a 104% rise in tensile strength, presenting the efficiency of symmetrical stacking. During flexural testing, the S2 configuration (G/Pv/G/Ph/G) recorded a 108% increase in strength as a result of the mixed orientation of paddy fibers increasing interlaminar bonding. Interestingly, the S4 configuration also recorded zero water absorption, highlighting its superior moisture resistance, while S5 (P/G/G/P) decreased water absorption by 10%.

These results confirm that strategic fiber stacking and nano-silica enhancement enhance mechanical strength and water resistance of agro-waste-based composites greatly. The optimum configuration not only uses renewable material but also holds excellent potential to be used in structural parts like two-wheeler visors, as in line with global trends towards light-weight, high-performance, and sustainable materials for the automotive, aerospace, and marine industries.

Table of Contents

CERTIFICATE *2*

ACKNOWLEDGEMENT *3*

Table of Contents **6**

List of Tables **8**

List of Figures *9*

CHAPTER 1: INTRODUCTION *10*

1.1 Composite material: **10**

1.2 Type of Composites: **12**

1.3 Type of Fibers: **13**

1.4 Type of Matrix **13**

1.5 Glass Fiber **14**

1.6 Key Features of the Composite: **14**

1.7Properties of Paddy Straw and Glass Fiber **15**

1.8 Application: **15**

1.9 Two Wheeler Wind Visor: **16**

CHAPTER 2: REVIEW OF LITERATURE *18*

SCOPE OF STUDY **20**

CHAPTER 3: OBJECTIVES OF THE STUDY *21*

RESEARCH METHODOLOGY *22*

WORK PLAN WITH TIMELINES *24*

CHAPTER 4: MATERIALS, METHODS & TESTING *25*

4.1: Material **25**

4.2: Methodology **27**

4.3: Methodology of Wind Visor: **30**

4.4: TESTING **32**

Tensile Test: 33

Flexure Test: 34

Water absorption Test: 35

CHAPTER 6: RESULTS & DISCUSSION ***37***

CONCLUSION ***51***

REFERENCE ***52***

APPENDIX ***55***

Review paper (communicated in Bulgarian Chemical Communications Journal) **55**

Patent under filing **57**

List of Tables

Table 1: Applications of Composites in various fields Page 15

Table 2: Physical Properties of Paddy Straw Page 25

Table 3: Mechanical Properties of E-glass Fiber Page Page 25

Table 4: Mechanical Properties of Epoxy Page 26

Table 5: Mechanical property of Nano silica Page 27

Table 6: Composite laminates (S1 to S6) without Nano silica Page 28

Table 7: Composite laminates (N1 to N6) with Nano silica Page 28

Table 8: Testing Sample Dimensions Page 34

Table 9 Tensile test results of without Nano Silica samples (S1-S6) Page 40

Table 10 Tensile test results of with Nano Silica samples (N1-N6) Page 43

Table 11 Flexure test results of without Nano Silica samples (S1-S6) Page 46

Table 12 Flexure test results of with Nano Silica samples (N1-N6) Page 50

List of Figures

Figure 1 Graphical schematic of the sample fabrication using hand-layup and compression molding technique 29
Figure 2 Neat Glass/Paddy reinforced epoxy composite specimen (S1-S6) 29
Figure 3 Sample (N1-N6) with Nano silica 30
Figure 4 Process of manufacturing wind-visor 31
Figure 5 Composite Laminate 33
Figure 6 Tensile Testing in UTM 34
Figure 7 Composite Laminate flexure test in UTM 35
Figure 8 Water absorption test sample 36
Figure 9: Tensile Stress-Strain of Sample S1 and S2 37
Figure 10 Tensile Stress-Strain of sample S2 and S3 38
Figure 11 Stress-Strain of Sample S5 and S6 39
Figure 12 Flexure Stress-Strain of sample N1 and N2 40
Figure 13 Flexure Stress-Strain of Sample N3 and N4 41
Figure 14 Flexure Stress-Strain of sample N5 and N6 42
Figure 15 Tensile Stress-Strain of sample S1 and S2 43
Figure 16 Tensile Stress-Strain of sample S3 and S4 44
Figure 17 Tensile Stress strain of sample S5 and S6 44
Figure 18 Flexure Stress-Strain of Sample N1 and N2 46
Figure 19 Flexure Stress-Strain of Sample N3 and N4 47
Figure 20 Flexure Stress-Strain of sample N5 and N6 47
Figure 21 Water absorption test results of sample without Nano silica (S1-S6) & with Nano silica (N1-N6) 50

CHAPTER 1: INTRODUCTION

Increased interest in sustainable materials has driven the development of hybrid composites that offer both environmental advantages and enhanced mechanical performance [2][6]. This research deals with the comparative evaluation of a glass fiber and paddy straw-reinforced hybrid composite in an epoxy matrix, highlighting its structural potential. By combining synthetic and natural reinforcement, a dual aim of improving material properties and sustainability is addressed by the research's focus [1][7].

Paddy straw is a by-product of rice production, and it tends to be wasted or burnt, which is a major source of pollution and greenhouse gas emissions [3][8]. Its loose, fibrous nature, as well as its ability to act as a thermal insulator, are ideal qualities for it to be used as a reinforcement in natural composites [4]. Glass fiber, however, is the epitome of tensile strength, rigidity, and heat resistance [2]. Blending these materials enables the development of a hybrid composite that utilizes the benefits of each with reduced environmental impact related to synthetic materials [5][9].

In the present research, the hybrid composite is produced by a manual layup method with paddy straw and glass fiber layers being bonded together with an epoxy resin [6]. Post-curing treatments are carried out to maximize the material's structural performance and thermal resistance [5][7]. A comparative analysis of mechanical properties like tensile strength, flexural strength, and impact resistance is done to determine the effect of different reinforcement ratios and layer arrangements

1.1 Composite material:

A composite material is a mixture of two or more different materials, the resulting material having properties better than the components. Typically, this consists of a matrix (usually a polymer, metal, or ceramic) which holds the structure together, and a reinforcement (e.g., fibers such as glass, carbon, or aramid) which adds strength and stiffness. The outcome is a material that is both strong but lightweight, being perfect for use in scenarios where one needs high

performance but with minimal weight [2][3]. The idea of composite existed as early as ancient days — the Egyptians, for instance, used straw combined with mud to construct bricks for building, which was an early composite. Yet, contemporary composite technology started developing in World War II with the introduction of fiberglass (glass fiber reinforced polymer) for military use. This provided the basis for the advanced composites that exist today in many different industries[1][4].

Composites are mostly applied in the automotive sector to minimize vehicle weight, resulting in better fuel economy and increased performance. Throughout the decades, materials like carbon fiber reinforced polymer (CFRP) and glass fiber reinforced polymer (GFRP) have made their way into automotive production in the form of body panels, bumpers, chassis parts, and interior components. CFRP is particularly applied in performance and high-end vehicles like McLaren, BMW, and Ferrari due to its excellent strength-to-weight ratio. GFRP, though costlier, finds extensive usage in mass-produced vehicles for applications such as fenders and hoods[3][5]. Also, certain environmentally friendly automobile makers are trying natural fiber composites, composed of plant-based material such as hemp or flax, to provide eco-friendly, light-weight interior elements[4].

Composites are still more vital in the aerospace sector as a result of the enormous focus on minimizing weight to optimize fuel efficiency and range of flight. Airplanes like the Boeing 787 Dreamliner and Airbus A350 use extensive amounts of their airframes made from advanced composites such as CFRP, which assist in minimizing overall weight while ensuring structural durability. Such materials also have good resistance to corrosion and fatigue, lowering maintenance expenses throughout the life of the aircraft [1][6]. Composites are also very commonly used in the production of aircraft wings, fuselage, and tail sections, and satellites and space vehicles for exploration where dependability in hostile environments is crucial.

One of the most common composites in both markets is FRP, or Fiber Reinforced Polymer. FRP composites consist of inserting durable, rigid fibers (such as glass, carbon, or aramid) into a lightweight polymer matrix (such as epoxy, polyester, or vinyl ester). This arrangement

enables engineers to design material properties for a variety of requirements, compromising between strength, flexibility, durability, and weight [2][7]. FRP composites are of a number of forms, with GFRP (Glass Fiber Reinforced Polymer) being the most abundant because it is inexpensive and has acceptable performance, whereas CFRP (Carbon Fiber Reinforced Polymer) is preferred in high-performance use because of its high strength and lightness[3][10]. AFRP (Aramid Fiber Reinforced Polymer), or commercially known as Kevlar, is applied where impact strength is important, e.g., in ballistic protection and certain applications in aerospace [6].

1.2 Type of Composites:

Composites are specially designed materials produced by bringing together two or more different substances to produce a material with properties better than those of the constituent parts. Depending on the nature of the matrix material used, composites are broadly categorized into polymer matrix composites (PMCs), metal matrix composites (MMCs), ceramic matrix composites (CMCs), and fiber-reinforced polymer (FRP) composites.

Polymer matrix composites, which use polymers like epoxy, polyester, or thermoplastics as the matrix, are most prevalent because of their low weight, corrosion resistance, and ease of manufacturing. They are commonly applied in the automotive market for body panels, bumpers, and dashboards, in aerospace for interior components and lightweight structures, and in sporting goods like tennis rackets, bicycle frames, and helmets.

Metal matrix composites are formed by combining metal matrices (e.g., aluminum or titanium) with reinforcements in the form of ceramic particles or fibers. The composites find application in aerospace, defense, and automotive technologies where high strength, thermal conductivity, and resistance to wear are required — for instance, in engine parts, brake systems, and structural frames.

Ceramic matrix composites provide superior thermal and chemical resistance and are well-suited to applications involving high temperatures. They include application in jet engines, rocket nozzles, turbine blades, and nuclear reactors, where materials have to endure high temperatures without deforming or degrading.

Fiber-reinforced polymer (FRP) composites are a particular and prevalent type of PMCs wherein strong fibers such as glass, carbon, or aramid are dispersed in a polymer matrix. FRP composites are particularly prized in civil engineering to strengthen bridges, buildings, and pipelines, in boat applications to boats and propellers, and even to wind turbine blades in renewable energy because they are lightweight with high fatigue endurance.

1.3 Type of Fibers:

Fibers employed in composites are commonly categorized as natural fibers (e.g., jute, hemp, coir, flax, and paddy straw), synthetic fibers (e.g., glass, carbon, and aramid), and mineral fibers (e.g., basalt). Each has distinct properties—natural fibers are light in weight, easily degradable, and inexpensive, whereas synthetic fibers have high tensile strength and resistance to chemicals. In our situation, glass fibers offer strength and structural integrity, while paddy straw fibers offer cost savings, eco-friendliness, and reduced weight to the composite.

1.4 Type of Matrix

The matrix in a composite material is the binding matrix that holds the fibers together, transfers loads among them, and shelters them from the environment. The performance and longevity of a composite are greatly determined by the matrix used. There are three principal types of matrix materials: polymer matrices, metal matrices, and ceramic matrices. Of these, polymer matrix composites (PMCs) are the most commonly employed because of their simplicity of manufacture, lightweight, and low cost. PMCs can be further subcategorized into thermosetting polymers (e.g., epoxy, polyester, and vinyl ester) and thermoplastics (e.g., polypropylene and polyethylene). Thermosetting resins are used in applications requiring high performance since they possess superior mechanical properties, resistance to chemicals, and stability of dimensions upon curing. Conversely, metal matrix composites (MMCs) that employ metals such as aluminum or magnesium as the matrix are utilized in applications where high thermal conductivity and structural strength are needed, for example, aerospace and automotive engine parts. Ceramic matrix composites (CMCs) are extremely heat resistant and are generally applied in harsh environments such as turbines and heat shields. In our research, we were interested in utilizing epoxy resin, a thermosetting polymer, as the matrix material because of its excellent mechanical strength, high adhesion to fibers, chemical resistance, and

high dimensional stability. This selection guaranteed that our composite developed using both glass fiber and paddy straw fiber as reinforcement had better performance with durability and sustainability.

1.5 Glass Fiber

Glass fiber, or fiberglass, is among the most common reinforcement material used in composite production because of its superior mechanical performance, affordability, and versatility. Glass fiber is composed of thin fibers of silica-based glass woven into mats, chopped, or produced in the form of continuous filaments. Glass fiber has high tensile strength, low weight, resistance to corrosion, and good insulation, which makes it a perfect material for various structural and non-structural uses. Glass fiber is utilized in the automotive sector for body panels, hoods, and bumpers to decrease the weight of the vehicle and improve fuel economy without sacrificing strength. In the aerospace industry, it is employed for secondary structure elements and interior, providing toughness and fire resistance. In shipbuilding, glass fiber is a favored material for boat hulls, decks, and masts due to its superior resistance to water and seawater corrosion. In civil engineering, glass fiber is employed to reinforce concrete, bridge decks, and structural retrofitting, considerably enhancing the toughness and load capacity of structures. Moreover, within the sports and recreation sector, it is employed in surfboard, ski, bicycle frame, and protective gear manufacturing, creating a balance of strength and elasticity. Because of its versatility and performance, glass fiber remains a core material of contemporary composite products in various industry

1.6 Key Features of the Composite:

- Sustainability: Minimizes pollution by recycling farm waste[1].
- Mechanical Strength: Merges the flexibility of paddy straw with the rigidity of glass fiber[2].
- Thermal Resistance: Protects from hot temperatures[4].
- Material Hardness: Provides toughness better than wood or PVC[6].

1.7Properties of Paddy Straw and Glass Fiber

Paddy straw is a natural fibrous material of low density, biodegradable, and having a large supply, especially in rice-growing areas. The fibers of paddy straw have insulating and natural reinforcement properties if they are integrated into a composite matrix [7]. Due to the exclusive properties of paddy straw like thermal insulation and low-cost fabrication, it can be an economically viable material to develop for sustainable applications [1].

Glass fiber, however, is a very versatile material, appreciated for its superior mechanical properties such as high tensile strength, rigidity, and thermal resistance [2][4]. Glass fiber in many industrial applications is used to improve the structural performance and thermal characteristics of composites [5]. Although glass fiber is synthetic and non-biodegradable, the fact that it is blended with paddy straw makes it possible to reduce total synthetic content without impairing material performance [6].

1.8 Application:

Table 1: Application of Composites in various fields

Sector	Application	Composites used	Key benefits
Automobile	Body parts, chassis, interiors	GFRP, CFRP, Natural fiber	Light weight, fuel efficiency, safety
Aerospace	Wings, interiors, UAVs	CRPF, Amarid thermoplastic	High strength, heat resistance, durability
Marine	Hulls, deck, propeller, vessels	GFRP, CFRP, hybrid composites	Corrosion resistance, reduce weight

Table 1 shows composites have been extensively applied in many fields because of their high strength-to-weight ratio, corrosion resistance, and durability. In the automotive sector, composites are employed in body panels, bumpers, hoods, and structural parts like drive shafts and leaf springs. These uses lower vehicle weight, increase fuel economy, and promote performance without a loss of strength. In the aerospace industry, composites are crucial in the construction of aircraft fuselage, wings, rotor blades, and interior components. Their low weight and fatigue-resistant capabilities make them suitable for enhancing fuel efficiency and

performance while capable of handling adverse environmental conditions. The ship-building industry is also greatly influenced by composites, mainly in ship hulls, decks, masts, and underwater fairings. These composites provide superior corrosion resistance, buoyancy, and durability and thus are used for long-term exposure in severe marine environments. In the recreational and sporting applications, composites find vast use in the production of tennis rackets, golf clubs, bicycles, helmets, snowboards, and surfboards, where their lightness and resistance to impact improve sportsman performance and equipment lifespan. Finally, in civil engineering, composites are utilized to strengthen bridges, columns, and beams with FRP rebars and wrapping systems. They also find application in facade panels, roof coverings, and water treatment equipment because they are resistant to chemicals, weathering, and corrosion, leading to longer lives and less maintenance. In all these industries, composites, particularly those with nano-silica added, are crucial in promoting material performance and design innovation

1.9 Two Wheeler Wind Visor:

A wind visor on a two-wheeler is a standard accessory for motorcycles and scooters, usually mounted above the headlamp. Its sole purpose is to redirect wind away from the rider, enhance aerodynamics, and provide slight protection from road debris and rain[1]. Though it plays a critical role in safety and comfort for riders, the material used to produce the majority of commercial wind visors presents several problems. The most used materials are polycarbonate (PC) and acrylonitrile butadiene styrene (ABS) both of which are petroleum-based synthetic plastics[2].

These materials, although lightweight and pliable, have a number of environmental and mechanical limitations. They are non-biodegradable to begin with. When they are discarded, they remain in the environment for centuries, adding to the mounting issue of plastic waste[3]. Additionally, PC and ABS production requires high energy consumption and carbon emission, with high impacts on the environment in terms of fossil fuel consumption and greenhouse gas emission[4].

Mechanically, both PC and ABS have poor resistance to UV radiation. Exposure to sunlight over time tends to cause yellowing, brittleness, and subsequent surface cracking[5]. Such alterations not only impair the visual appearance of the visor but also diminish its performance and longevity. Moreover, these materials are also susceptible to surface scratches, which can distort vision and detract from safety while riding[6]. At elevated temperatures, they soften or deform, which affects structural integrity[7].

Another major drawback is their poor recyclability. The infrastructure and cost associated with recycling these plastics are often not feasible, which results in most used visors being dumped in landfills or incinerated both of which are environmentally harmful[8].

To counter these challenges, we suggest a glass fibre-reinforced composite from paddy straw agro-waste and epoxy resin as an eco-friendly replacement[9]. Not only does this new composite recycle agricultural waste that is normally incinerated and leads to air pollution, but it also improves the mechanical strength of the visor[10]. Our composite has higher strength, impact resistance, UV stability, and thermal resistance than traditional plastic materials.

Furthermore, the weight-to-strength ratio of this composite is greater, indicating that it is able to provide greater performance with less material, resulting in a reduced overall weight for the vehicle and better fuel efficiency[12]. It is more resistant, less susceptible to environmental degradation, and built to last longer than standard plastic visors[13].

CHAPTER 2: REVIEW OF LITERATURE

Maraveas, C. et al. (2018): examined the use of agro-waste in sustainable building materials. The research examined different methodologies for building material and composite manufacturing, including the use of particle boards and thermal insulated panels with various agro-waste material ratios. The review established that recycling unregulated agro-waste into sustainable building materials provides an effective route to pollution abatement and natural resource preservation.[16]

K. Vinod, S. Das. et al (2019) Synthesized hybrid aluminum matrix composites by reinforcing Al–7Si–0.3Mg alloy with agro-industrial waste materials. The tribological behavior of the composites was investigated, and it was found that the addition of waste particles enhanced wear resistance and hardness. The work demonstrated the possibility of utilizing agricultural waste as a reinforcement material in metal matrix composites for improved mechanical properties. SpringerLink[17]

V. Dwivedi et al (2019) Explored the physico-chemical, mechanical, and thermal properties of RHA green composites. The research showed that the addition of RHA in polymer matrices increased the mechanical properties and thermal stability of the composites, thus making them eligible for sustainable engineering purposes. SpringerLink[18]

O. P. Durowaye. et al (2019) Studied the influence of agro-waste particles, such as coconut shell, periwinkle shell, and cow bone, on the mechanical properties of hybrid unsaturated polyester resin matrix composites. The study revealed that 15 wt.% reinforcement gave the maximum mechanical properties, showing the effectiveness of using agro-waste particles to reinforce composite materials. Nepal Journals Online[19]

M. P. Soh et al (2020) Synthesized an extrudable paste to construct mycelium-bound composites from agricultural waste materials such as bamboo microfibers. The research sought to produce environmentally friendly materials to be used in packaging, insulation, fashion, and architecture. The produced composites had viable mechanical properties and design versatility. arXiv [20]

P. Raut et al (2012) Discussed the use of agro-waste for sustainable building materials. The research presented different methods of manufacturing building materials and composites, including particle boards and thermal insulated panels, from different ratios of agro-waste materials. The research concluded that recycling agro-waste into building materials provides a sustainable solution to pollution control and natural resource conservation.[21]

T. Kılıç et al (2024) Created environmentally friendly brake friction composites based on rice husk and rice stalk wastes. Through preparation of composites with 5% and 10% weight fractions of these agricultural wastes, the research evaluated their physical, mechanical, and thermal characteristics. The results indicated that composites with 5% rice husk had a good coefficient of friction and enhanced fade and recovery behavior, indicating their use in sustainable brake pad applications.[22]

L. Zhegur-Karady et al (2019) Investigated the application of wheat straw fibers, which are obtained from agricultural residues, as a reinforcement in wood plastic composites (WPC). They utilized steam explosion and chemical surface modification methods to improve the compatibility of wheat straw fibers with polypropylene matrices. The treated fibers resulted in improved thermal stability and mechanical properties, including stiffness and toughness, proving the potential of wheat straw as a renewable reinforcement in WPCs.[23]

H. Zhao et al (2022) Examined the use of peanut shells, a farm waste, as a green filler in polyamide bio composites. The work entailed preparation and characterization of biocomposite packaging films and determination of their mechanical and barrier properties. Addition of peanut shell powder enhanced the tensile strength and lowered the water vapor permeability of the films, signifying their prospect in green packaging applications.[24]

Laad et al (2018) Investigated the use of rice husk ash, which is an agricultural waste, as a reinforcing material in aluminum metal matrix composites (AMMCs). The experiment entailed the production of composites through stir casting methods and their mechanical properties testing. Adding 3% rice husk ash by weight enhanced the hardness and Young's modulus of the aluminum matrix, indicating the viability of using agricultural waste in strengthening the performance of metal composites.[25]

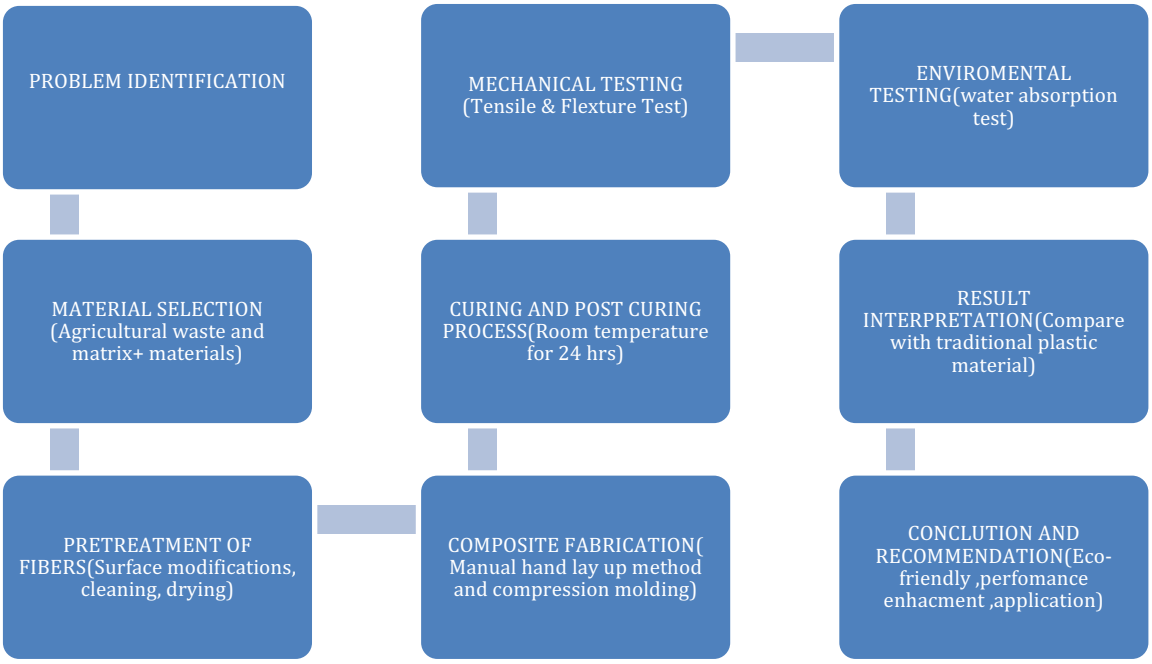
SCOPE OF STUDY

- To fabricate sustainable glass/paddy straw reinforced epoxy composites using hand lay-up and compression molding techniques.
- To evaluate the tensile strength, flexural strength, of the developed composites under varying material (fiber content, orientation, layering) and geometrical parameters.
- To assess the water absorption behavior of the composites under standardized testing conditions, analyzing their suitability for semi-structural applications in humid environments.
- To design and develop a two-wheeler wind visor using the best-performing composite configuration.

CHAPTER 3: OBJECTIVES OF THE STUDY

- Fabrication of laminate made from Glass fibre, Paddy, straw reinforced with Epoxy composites.
- Fabrication of visor with the same material (Glass fibre, Paddy Straw, and Epoxy).
- Mechanical characterization of Glass/Paddy Straw reinforced Epoxy composites (Tensile, Flexure)
- To evaluate the water absorption test of various configurations based specimens.

RESEARCH METHODOLOGY



- **Material Selection**

The primary materials selected for composite fabrication include glass fiber (E-glass), paddy straw (processed and treated), epoxy resin (as the matrix), and suitable Nanoparticles (e.g., Nano-silica or Nanoclay) as secondary fillers to enhance mechanical and barrier properties.

- **Specimen Fabrication**

Composite specimens were fabricated in various configurations by varying fiber volume fractions, layering sequences, and Nanoparticle loadings. Two fabrication techniques were employed:

- **Hand lay-up technique** for preliminary and small-scale laminates.

- **Compression molding** for better compaction and uniformity in large or complex-shaped parts. The fabricated specimens were cured under controlled conditions before testing.

- **Mechanical and Physical Characterization**

The fabricated specimens underwent a series of tests in accordance with ASTM standards to evaluate their mechanical and physical properties:

- **Tensile Test** (ASTM D3039)
- **Flexural Test** (ASTM D790)
- **Water Absorption Test** (ASTM D570)

- **Result Analysis and Configuration Optimization**

The mechanical performance and water absorption characteristics of different configurations were analyzed to identify the optimal combination that offers maximum strength and minimum water uptake. Statistical analysis and comparative evaluation were conducted to finalize the best-performing configuration.

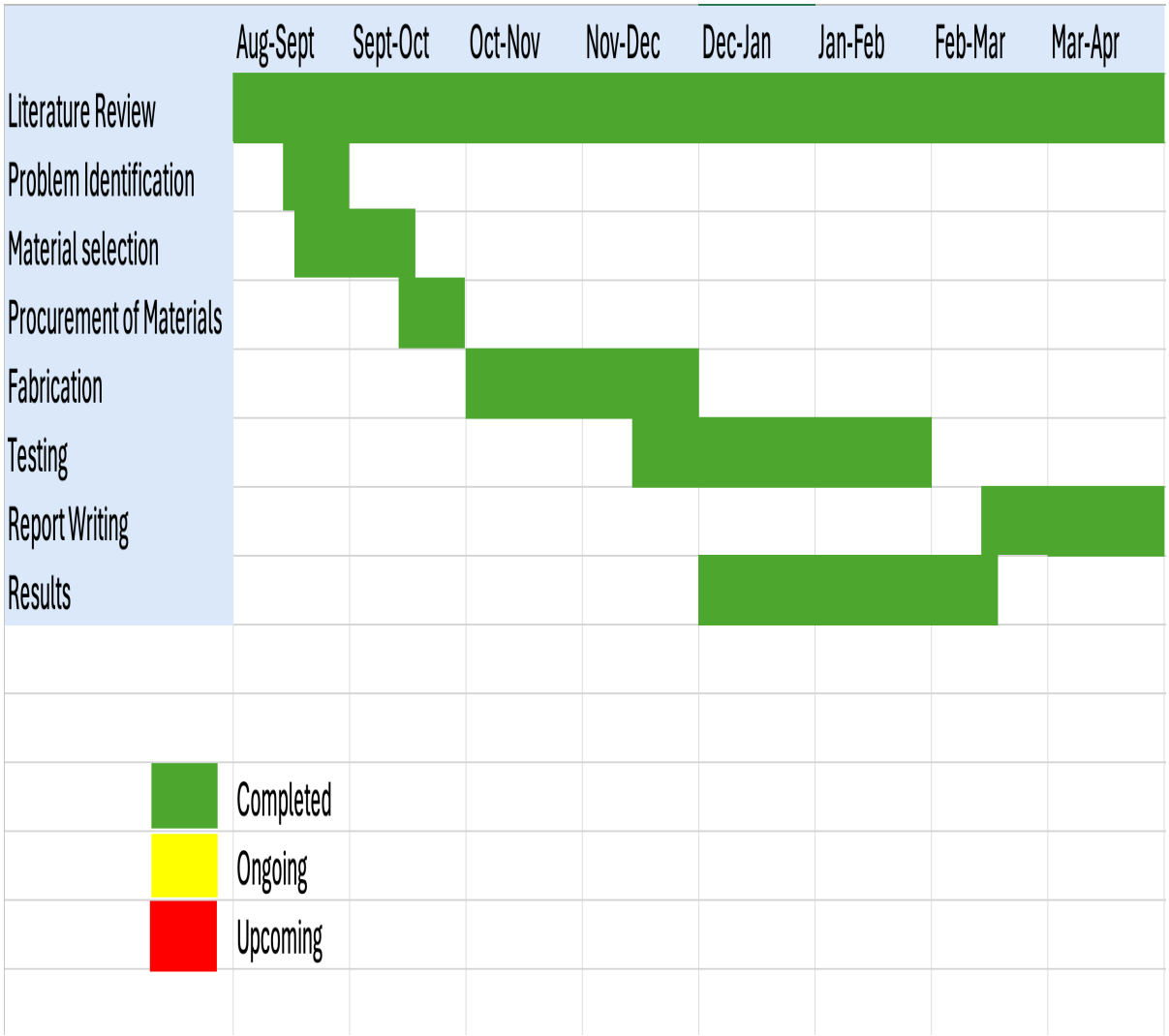
- **Mold Design and Prototype Fabrication**

A pattern of the two-wheeler wind visor (TVS Jupiter 125cc) was used to prepare a POP mold (cope and drag) suitable for composite lay-up. Using the optimized composite configuration, prototype wind visors were fabricated via both hand lay-up and compression molding techniques to assess real-world applicability and form consistency.

- **Documentation and Dissemination**

A detailed project report was prepared documenting all stages of the work, including methodology, results, and discussions. Further, research papers are being drafted for submission to reputed journals or conferences in the field of composite materials and sustainable engineering.

WORK PLAN WITH TIMELINES



CHAPTER 4: MATERIALS, METHODS & TESTING

4.1: Material

Burning Agro-waste, e.g., paddy straw, causes air pollution by emitting harmful gases and particulate matter. Nevertheless, it can be reused as a secondary raw material to produce composite materials towards sustainable development. Paddy straw possesses desirable physical characteristics such as low density, chemical components such as cellulose, hemicellulose, and lignin, and sufficient mechanical strength shown in Table 2 , which renders it useful as reinforcement in composite production. At first, the paddy waste was treated using an alkaline solution to enhance the adhesion. Then, the paddy was cut and laminated together to create a ply, which was employed to fabricate a glass/paddy-epoxy hybrid composite.

Table 2 Physical Properties of Paddy straw

Density	450 kg/m3
Water absorption	high water absorption rate, reaching 234%
Composition	35–40% cellulose, 25–30% hemicellulose, 10–15% lignin, 6–12% silica
Specific weight	70–80 kg/m3

The E-glass plain weave cloth with a weight of 290 ± 10 GSM was purchased from Navbharat Agency in Jalandhar, Punjab, India, and utilized as reinforcement. It possesses excellent tensile strength, lightweight nature, excellent electrical insulation, corrosion and chemical resistance, temperature and size stability, and economy. Mechanical Property of E-glass Fiber shown in Table 3

Table 3 Mechanical Properties of E-glass Fiber

Density	2540 kg/m³
Tensile strength	3.4GPa
Modulus of elasticity	72 GPa

Percent elongation	4.7
Shear modulus	106 psi
Glass transition temperature	840 °C

Table 4 Mechanical Properties of Epoxy

Property	**Epoxy LY556**	**Hardner HY951**
Visual appearance	Medium viscosity, colorless clear liquid	Brownish yellow color liquid
Viscosity at room temperature	9000-12000 MPa	500-1000MPa
Density at room temperature	1.13-1.16 gm/cc	0.946 gm/cc

Epoxy was manufactured by Huntsman Araldite resin LY551 and hardener HY951, which were obtained from CF Composite, New Delhi, India, and the mixing ratio is 10:1. Epoxy is one of the best of all polymers because of its outstanding mechanical properties shown in Table 4 , high adhesive strength, and good chemical resistance. It offers high tensile strength, rigidity, and heat stability, and hence it is applicable for structural purposes.

We have incorporated nano-silica in manufacturing our FRP (Fiber Reinforced Polymer) composite, which we source from Intelligent Materials in Mahali, Punjab. The particle size of the nano-silica is 10 nanometers, which is a critical factor in improving the overall properties of the composite. Our composite contains paddy, glass fiber, epoxy, and nano-silica, and all these individual elements support the final material performance. Nano-silica, with its very fine particle size, enhances the surface area of the composite, facilitating enhanced bonding between the glass fibers and the epoxy resin. This enhances the mechanical properties like tensile strength, impact resistance, and stiffness of the composite property of nano-silica shown in Table 5. Nano-silica also increases the wear and abrasion resistance, making the composite more resistant to harsh conditions. The addition of nano-silica also assists in minimizing creep behavior of the material, which makes it stable for the long term under load. Its reinforcement at the Nano level offers enhanced toughness and resistance to cracks, which are essential for FRP composites used in a wide range of applications.

Table 5 Mechanical property of Nano silica

Tensile strength	10-50MPa
Flexture strength	50-150Mpa
Hardness	7 (Mohs scale)

4.2: Methodology

The production of composite laminates for this project was a systematic and well-controlled process to test the efficacy of agro-waste-based reinforcement, in this case, paddy straw, in combination with glass fiber and epoxy resin. The process shown in Figure 1 started with the pre-treatment of paddy straw, where the fibers were subjected to alkaline chemical treatment using a sodium hydroxide (NaOH) solution to eliminate unwanted materials like lignin, hemicellulose, and waxes. This treatment enhanced the bonding capability of the straw with the matrix by increasing surface roughness and decreasing hydrophilicity. The treated fibers were thoroughly washed with distilled water to remove any residual chemicals and then oven-dried to eliminate remaining moisture.

After treatment, paddy straw and glass fibers were trimmed into accurate sizes of 10.5 cm × 20 cm (with tolerance ±5 mm) to ensure uniformity in lay-up. The weight of each layer of paddy straw was around 11.75 grams to ensure uniform material distribution throughout samples. The hand lay-up method was employed for the fabrication of laminate because it was flexible, cost-effective, and allowed easy layer control. Contrary to conventional molds, Teflon sheets were applied to both sides of the laminate in order to avoid sticking during curing and to aid in the formation of a smooth surface finish. The composite layers, which comprised mixes of paddy straw, glass fiber, and epoxy resin (with hardener), were stacked manually in accordance with pre-specified stacking sequences. Following lay-up, the laminate was introduced into a compression molding machine where it was exposed to a pressure and temperature of 90°C for 30 minutes. This enabled proper resin flow, air gap elimination, and compaction of the laminate. The compressed samples were next cooled under pressure for 30 minutes before post-curing at room temperature for 24 hours to finalize the hardening of the resin. 12 laminate samples were prepared by the Taguchi design of experiments method. They

were split into two groups: Group 1: Without Nano Silica (Sample Names: S1 to S6) shown in table 6 , Group 2: Nanoparticles added as 2% of the total weight (paddy + glass + epoxy) (Sample Names: N1 to N6) shown in table 7 . Three specimens of each configuration were fabricated to obtain the average strength.

Example for Nanoparticle addition:

- Total weight of paddy + glass = 40 g.
- Epoxy weight = 110 g.
- Total = 40+110= 150 g.
- Nanoparticles = 150×0.02= 3 g

Table 6 Composite laminates (S1 to S6) without Nano silica

Specimen no	Stacking Sequence	Thickness (mm)	Filler material (wt.%)
S1	GPGPG	2.18±0.15	0%
S2	GPGPG	2.15±0.15	0%
S3	GPPG	1.98±0.15	0%
S4	GGGG	1.73±0.15	0%
S5	PGGP	2.18±0.15	0%
S6	PPPP	4.2±0.15	0%
***G = Glass fiber, *P = Paddy straw**			

Table 7 Composite laminates (N1 to N6) with Nano silica

Specimen no	Stacking Sequence	Thickness (mm)	Filler material (wt.%)
N1	GPGPG	2.11±0.15	2%
N2	GPGPG	1.85±0.15	2%
N3	GPPG	2.08±0.15	2%
N4	GGGG	1.07±0.15	2%
N5	PGGP	2.06±0.15	2%
N6	PPPP	3.06±0.15	2%
***G = Glass fiber, *P = Paddy straw**			

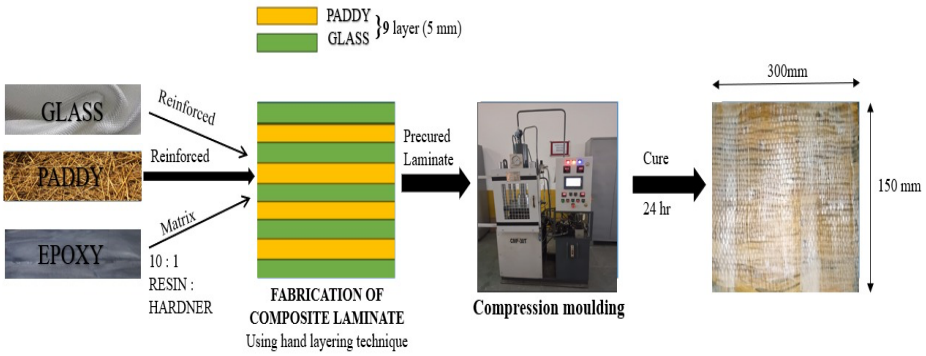


Figure 1 Graphical schematic of the sample fabrication using hand-layup and compression molding technique

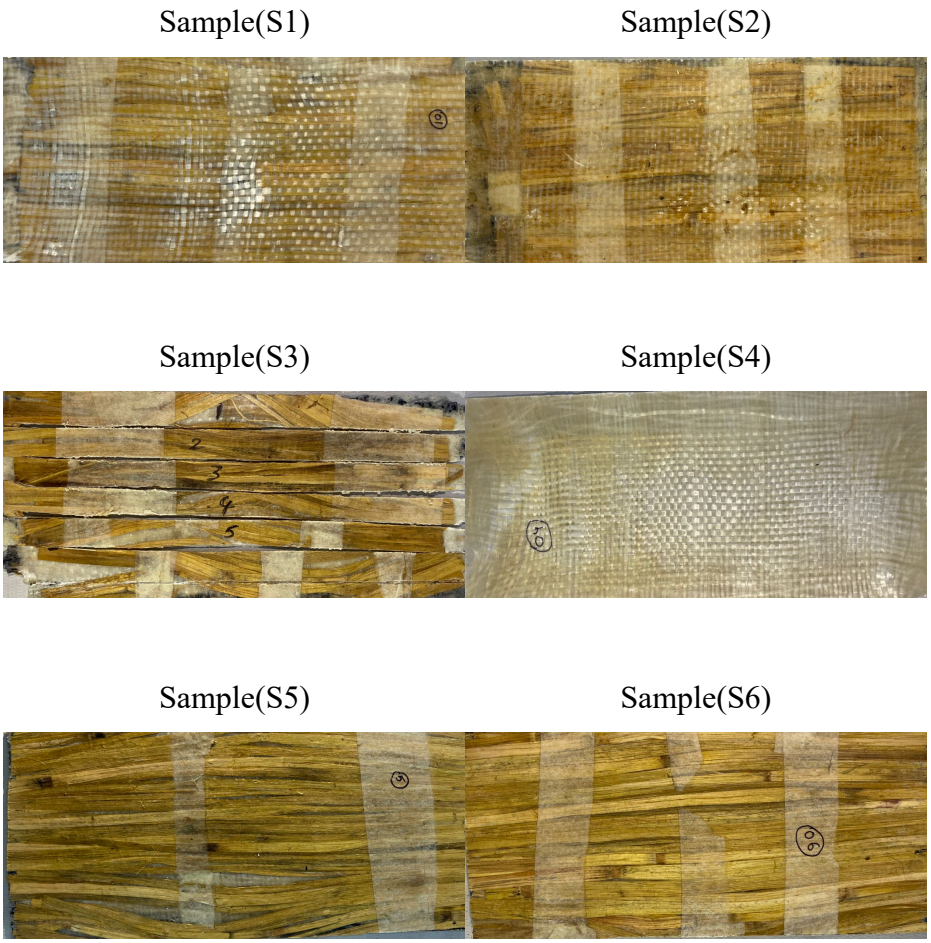


Figure 2 Neat Glass/Paddy reinforced epoxy composite specimen (S1-S6)

Sample(N1)



Sample(N2)



Sample(N3)



Sample(N4)



Sample(N5)



Sample (N6)

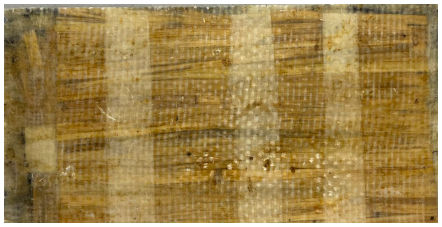


Figure 3 Sample (N1-N6) with Nano silica

4.3: Methodology of Wind Visor:

The manufacturing of two-wheeler wind visor from a composite of paddy straw, glass fiber, and epoxy resin shown in Figure 4 was done through a well-controlled process making use of both the manual lay-up and compression molding methods. The process started with the development of a mold, which was produced by utilizing a conventional cope and drag arrangement prepared from Plaster of Paris (POP). This two-piece mold gave an exact cavity that defined the ultimate shape and surface detail of the visor.

First, the natural reinforcement material, paddy straw, was gathered, washed to purify it from impurities, and chopped into short fibers. These were alkaline treated using a 5% sodium hydroxide (NaOH) solution for approximately two hours in order to enhance interfacial adhesion with the epoxy resin. The treated fibers were then washed extensively using distilled water and dried in an oven to eliminate moisture. In addition, glass fiber mats were trimmed to the mold dimensions to be the secondary reinforcement, providing structural stability and enhancing surface finish. The hardener and epoxy resin were blended in a generic 10:1 ratio to be the matrix material for the binding of the fibers. After the materials were prepared, the fabrication process started with the placement of the drag half of the mold firmly. A glass fiber layer was initially placed on the drag mold, followed by an alkali-treated paddy straw layer, and then epoxy resin-impregnated using a brush to provide even wetting and bonding.



Figure 4 Process of manufacturing wind-visor

This process was repeated until the desired thickness was obtained. Once the lay-up was done, the cope half of the mold was placed on top to close the mold assembly. To provide satisfactory compaction and adequate fiber-resin bonding, the whole assembly of the mold was subjected to a compression-molding machine. An evenly distributed load of 2 kN across the surface pressed the composite layers in a compression mode. This pressure eliminated air bubbles, minimized voids, and provided high fiber volume fraction. The compressive load kept the mold at room temperature for 24 hours to facilitate curing of the resin. Once cured, the visor was carefully demolded and underwent finishing operations like trimming and sanding to achieve the final shape and a smooth finish.

4.4: TESTING

For mechanical and physical testing of the FRP laminates, the following was done:

1. **Sample Preparation**: The laminates were cautiously cut as per ASTM composite standards for testing in order to produce uniformity and consistency in test samples shown in Figure 5.
2. **Sample Selection**: Three samples for each laminate type were taken in order to achieve accurate and dependable results. The mean value of the three results was obtained in order to derive a more accurate outcome.
3. **Tests Performed**: The samples were subjected to the following tests:
    - Tensile Test: To assess the tensile strength and measure the material's resistance against stretching pull.
    - Flexural Test: To determine the flexural strength and material's capacity to resist bending or deformation under load.
    - Water Absorption Test: To determine the extent to which moisture is absorbed by the composite, which can impact its durability and long-term performance.



Figure 5 Composite Laminate

Testing Samples Dimensions (cut after block preparation):

Table 8 Testing Sample Dimensions

Test Type	Standard	Dimensions
Tensile Test	ASTM D3039	250 × 15 mm
Flexural Test	ASTM D790	125 × 13 mm
Water Absorption Test	ASTM D570	20 × 20 mm

Tensile Test:

1.Definition of Terms:

Tensile Strength: Maximum stress a material can withstand before fracture when stretched.

Yield Strength: Stress at which the material begins to deform permanently.

Ductility: The ability of the material to stretch without fracture.

2.Testing Standard: ASTM D3039 (Standard for composite materials).

3.Sample Preparation:

Sample was cut to dimensions 250mm x 15mm.

Three samples were tested for precision.

Average of three results was considered as final value.

4.Testing Equipment: Universal Testing Machine (UTM) shown in Figure 6 .

5.Test Parameters:

UTM speed at 2mm/min.

Maximum load applied: 900kg.

6.Function: The test assesses how effectively the material resists separation, guaranteeing its strength and performance under tension.



Figure 6 Tensile Testing in UTM

Flexure Test:

1. **Definition of Key Terms**:

   Flexural Strength: Material's resistance to bending before fracture.

   Stiffness: Material's resistance to bending deformation.

   Load-Carrying Capacity: Maximum load material can carry before failing.

2. **Testing Standards**:

   ASTM D7264

   ASTM D790

3. **Sample Preparation**:

   Samples were cut into 250mm x 15mm size.

Three samples were tested for consistency.

Average of three readings was considered as final reading.

4. **Testing Equipment**: Universal Testing Machine (UTM) shown in Figure 7.
5. **Test Parameters**:

   UTM speed adjusted at 5mm/min.

   Maximum load applied: 225kg.

6. **Purpose:**

   Determines the material's bending resistance and structural integrity under load conditions.



Figure 7 Composite Laminate flexure test in UTM

Water absorption Test:

1 **ASTM Standard**: ASTM D570 (Standard Test Method for Water Absorption of Composites).

2 **Purpose**: Specifies the water absorbed by the composite material in a given time.

3 **Sample Dimensions**: Usually 76.2mm × 25.4mm × 3.2mm (3 inches × 1 inch × 0.125 inches) shown in Figure 8 , but depending on material.

4 **Procedure**:

Dry the specimen in an oven at 50-60°C until constant weight.

Immerse the specimen in distilled water at 23°C (±2°C) for 24 hours.

Remove and wipe the surface to clean off excess water.

Weigh the specimen immediately to get the absorbed water content.

5 **Result Calculation**:

Water absorption (%) = ((Wet weight - Dry weight) / Dry weight) × 100.

6 **Significance**:

Aids in evaluating moisture resistance, durability, and fit of the composite into various environments.



Figure 8 Water absorption test sample

CHAPTER 6: RESULTS & DISCUSSION

Tensile Test without Nano-Silica

Following is a critical review of the tensile test data for each of the composite laminate specimens (S1 to S6):

This laminate, alternating between glass fiber and paddy straw, exhibits a relatively moderate peak load of 145.7 kg and tensile strength of 53.305 N/mm². Its balanced stacking causes relatively even stress distribution. The tensile strength, though, is below anticipated levels for a hybrid laminate with glass fiber, indicating that paddy straw, although light, is possibly undermining the load-carrying capability of the laminate. The elongation of 4.65 mm indicates moderate ductility. While having a somewhat different orientation in paddy straw placement than S1, the performance of S2 is nearly identical to S1, with a tensile strength of 51.784 N/mm² and maximum load of 143 kg. The small reduction may reflect a slight loss in interlaminar bonding caused by fiber misalignment. The elongation of 5.14 mm indicates superior strain absorption compared to S1, but no notable mechanical benefit is provided through vertical-horizontal orientation shown in Figure-9.

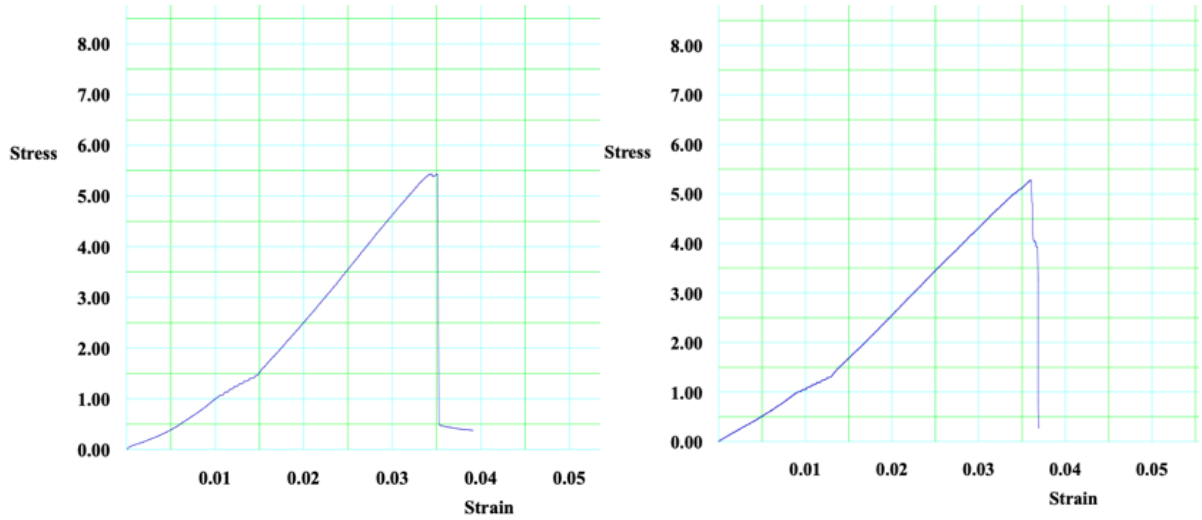


Figure 9 Tensile Stress-Strain plot of sample S1 and S2

This stacking, with glass-paddy-paddy-glass orientation, shows a significant reduction in strength, 104.5 kg maximum load and 39.827 N/mm² tensile strength. The lower glass fiber content and more central location of the weaker paddy layers probably reduced the strength. The tensile performance is significantly less than S1 and S2, highlighting the structural significance of glass fibers in both outer and inner layers. The elongation of 4.1 mm also shows somewhat brittle behavior. Contrary to expectations, the all-glass fiber laminate has the least tensile strength. With maximum load of only 79.6 kg and tensile strength of 42.582 N/mm², there is apparent bad interlaminar bonding or defects of the manufacturing type (voids, resin rich region) occurring within the test pieces. Since the glass-only laminate would naturally have been predicted to perform better than the hybrids, but registers a lowest elongation value (3.16 mm) inferring that it has failed in more brittle mode than others have shown in Figure-10.

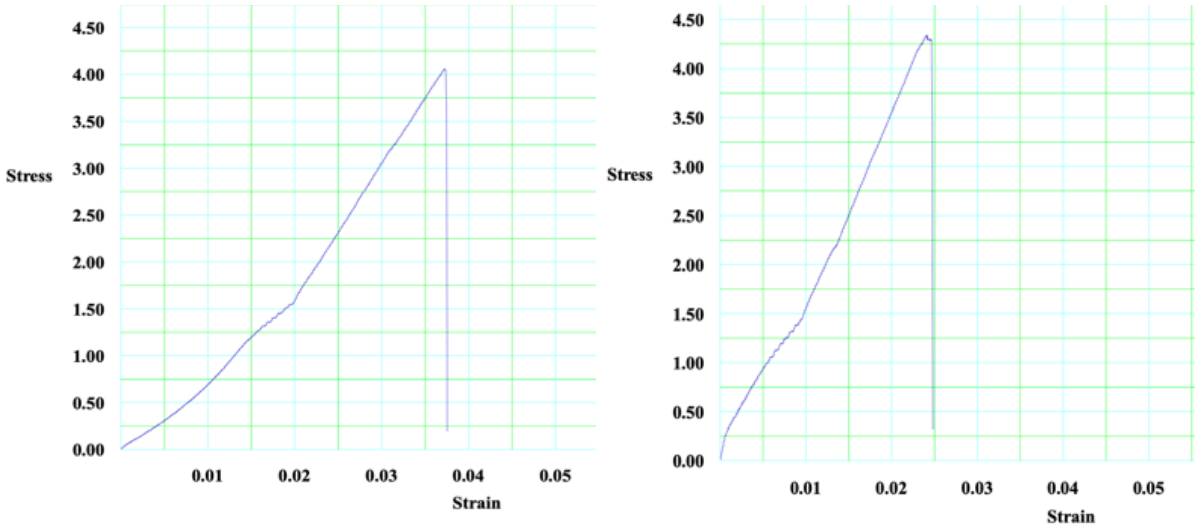


Figure 10 Tensile Stress-Strain plot of sample S3 and S4

This laminate displays the optimal composite tensile behavior, with peak load 187.7 kg and tensile strength 63.508 N/mm². The likely more favorable stacking arrangement may ensure better stress transfer and inter-layer bonding. Elongation value of 5.46 mm is also indicative of a strong to ductile property balance. This demonstrates that proper configuration is achievable, where hybrid laminates incorporating agro-waste reinforcement could be better than conventional ones. As predicted, the all-paddy laminate acts the poorest regarding

strength-to-thickness ratio. Even though the maximum load is great (187.6 kg) because of higher thickness (4.2 mm), the tensile strength falls to a mere 36.312 N/mm², the lowest of all. The bigger cross-section took up more load, but poor matrix-fiber bonding in paddy-only stacks impacted the performance. But the peak elongation (6.83 mm) indicates that this sample extended significantly before fracture, implying greater flexibility but extremely poor strength shown in Figure-11.

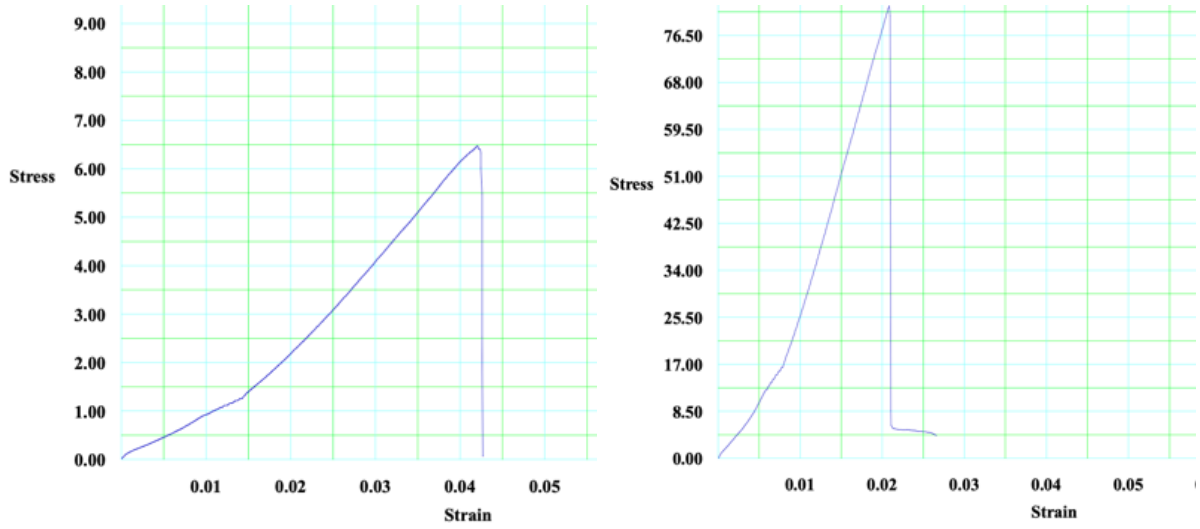


Figure 11 Stress-Strain plot of sample S5 and S6

Table 9: Tensile test results of without Nano Silica samples (S1-S6)

Specimen no	Stacking Sequence	Thickness (mm)	Peak Load (kg)	Tensile strength (MPa)
S1	GPGPG	2.18±0.15	145.7	53.305
S2	GPGPG	2.15±0.15	143	51.784
S3	GPPG	1.98±0.15	104.5	39.827
S4	GGGG	1.73±0.15	79.6	42.582
S5	PGGP	2.18±0.15	187.7	63.508
S6	PPPP	4.2±0.15	187.6	36.312

Tensile Test with Nano-Silica

The following is a critical comparison and analysis of the tensile test results of the Nano-silica shown in Table 10 reinforced composite laminates (N1 to N6) versus their non-Nano counterparts (S1 to S6), including percentage improvement in tensile strength:

In comparison with S1, tensile strength showed an increase from 53.305 N/mm² to 80.3316 N/mm², which is a ~50.7% increment. This strong improvement verifies the reinforcing role played by Nano silica, which would have increased the bond between fibers and matrix, enhancing stress transfer. The elongation also showed a slight improvement to 5.21 mm, reflecting improved flexibility as well as strength, thereby forming a well-balanced composite. This sample demonstrated a rise in tensile strength from 51.784 N/mm² to 58.939 N/mm², a ~13.8% increase. While less spectacular than N1, the increase confirms that Nano silica continues to play a beneficial role. The small increase in elongation to 5.24 mm confirms improved energy absorption, though the variable stacking orientation may have prevented further increases shown in Figure-12.

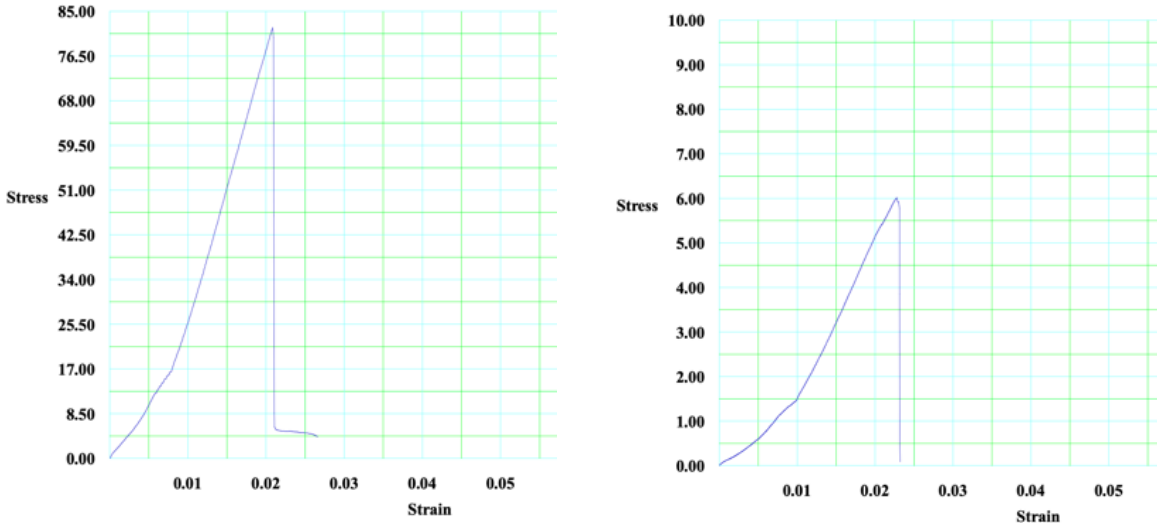


Figure 12 Flexure Stress-Strain plot of sample N1 and N2

This laminate experienced a dramatic increase in tensile strength from 39.827 N/mm² to 81.3098 N/mm², that is, a 104% or more than a 104% increase. This tremendous improvement signifies that Nano silica significantly mitigates the inner weakness due to inner paddy layers. Elongation was also increased to 5.71 mm, indicating that Nano-modified matrix significantly improved both ductility and strength. This specimen exhibits the highest tensile strength of all:

127.1909 N/mm², as opposed to 42.582 N/mm² in its non-Nano equivalent—a whopping ~198.7% improvement. The complete glass fiber stacking with Nano silica produced a very strong composite. Elongation was still low at 3.99 mm, however, with the material still being brittle, though extremely strong. Nano silica greatly enhanced interfacial adhesion in this brittle system shown in Figure-13.

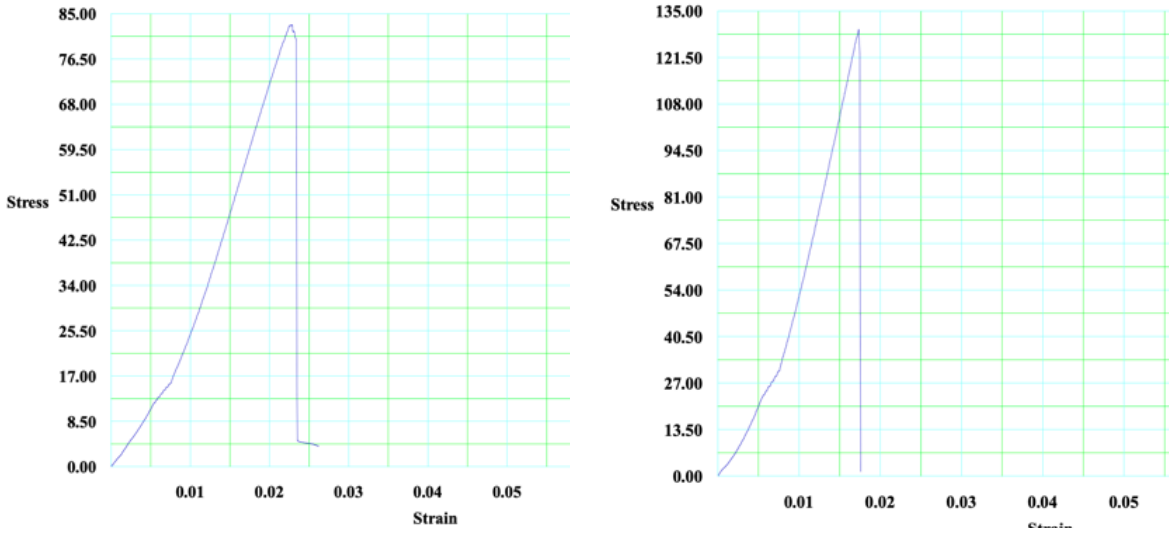


Figure 13 Flexure Stress-Strain plot of sample N3 and N4

The tensile strength was increased from 63.508 N/mm² to 76.8367 N/mm², an increase of ~21%. This sample was already among the top performers in the non-Nano group, and better bonding helped it. The elongation increased to 6.45 mm, which indicates enhanced flexibility and toughness and is a sought-after property in structural usage. Here, tensile strength improved from 36.312 N/mm² to 48.489 N/mm², a ~33.5% improvement, which is notable considering it's a natural fiber-only laminate. The elongation was 8.09 mm, the best among all, showing great deformability. But even though Nano silica improves bonding in weak matrices such as paddy straw, the lack of glass restricts the strength potential. Nano silica exhibits a uniformly beneficial effect in all stacking sequences, with the most substantial gains in GGGG (N4) and GPPG (N3) laminates. It increases tensile strength by optimizing matrix-fiber interaction and efficiency of stress transfer. The influence is more intensified in laminates where previously, fiber-matrix bonding was a soft spot. The drawback is merely that while

there is strength gain, there still exists brittleness in neat glass laminates such as N4 with low elongation shown in Figure-14.

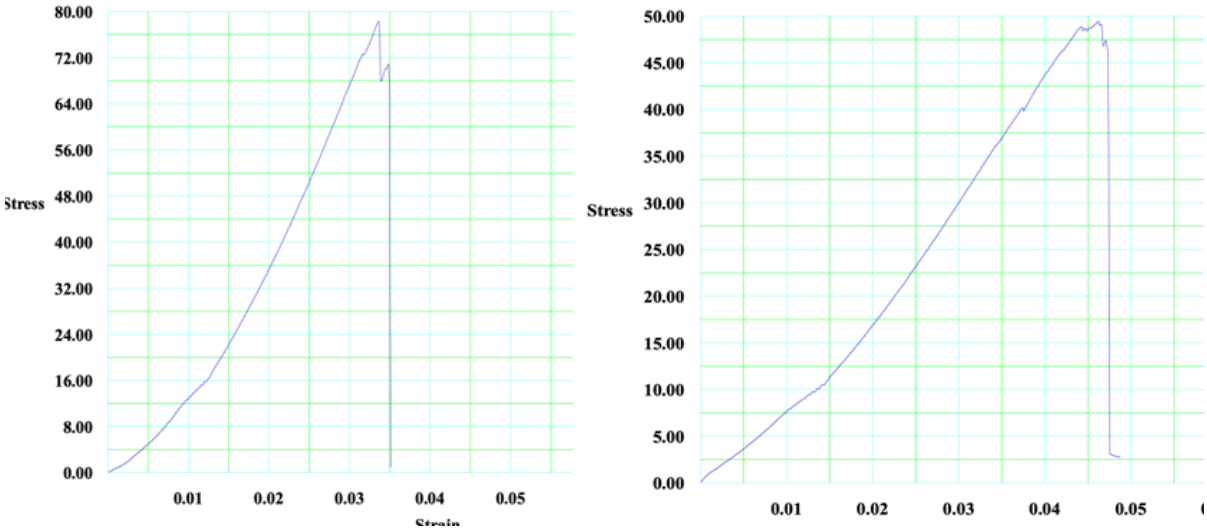


Figure 14 Flexure Stress-Strain plot of sample N5 and N6

Table 10 Tensile test results of with Nano Silica samples (N1-N6)

Specimen no	Stacking Sequence	Thickness (mm)	Peak Load (kg)	Tensile strength (MPa)
N1	GPGPG	2.11±0.15	198.7	80.3316
N2	GPGPG	1.85±0.15	165	58.939
N3	GPPG	2.08±0.15	215.5	81.3098
N4	GGGG	1.07±0.15	174.8	127.1909
N5	PGGP	2.06±0.15	203.3	76.8367
N6	PPPP	3.06±0.15	181.5	48.489

Flexure Test without Nano-Silica

The critical analysis of the flexural test results for the composite laminates (S1–S6) without Nano silica shown in Table 11 is presented below in terms of their stacking sequence, load response, flexural strength, and deformation behavior:

This laminate exhibited a moderate flexural strength of 88.456 N/mm² and deflection of 4.56 mm, exhibiting a balanced mix of flexibility and strength. The alternating glass and paddy straw layers probably achieved good load distribution, but the straw may have slightly curbed optimum performance. It overall exhibits stable behavior when subjected to flexural stress. Having a lower flexural strength of 47.025 N/mm² and deflection of 5.05 mm, S2 performs below S1 despite both containing the same constituents. The fiber orientation variation (horizontal and vertical) potentially caused an inconsistent stress transfer, thereby becoming weaker in bending loads. Stacking strategy at the cost of mechanical synergy shown in Figure-15.

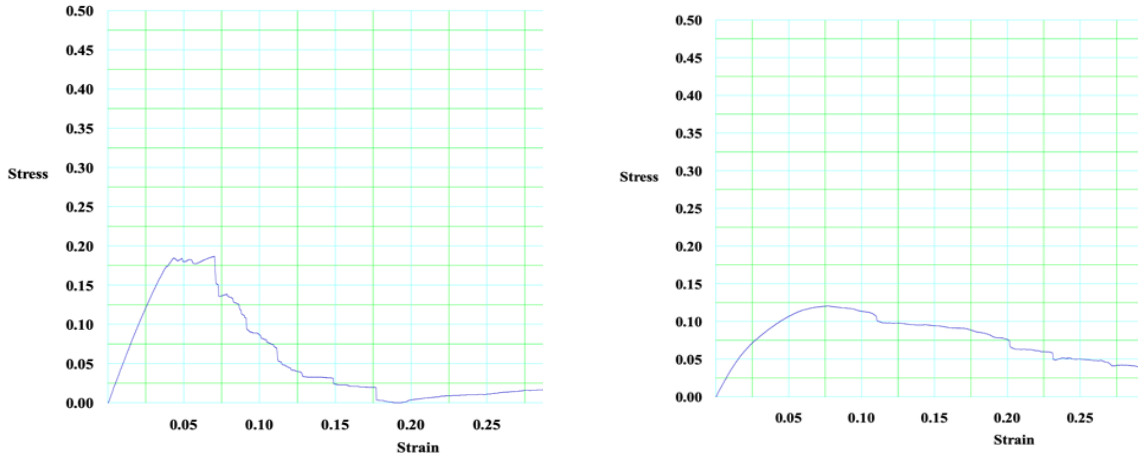


Figure 15 Flexure Stress-Strain plot of sample S1 and S2

S3 showed a flexural strength of 74.447 N/mm², inferior to S2 but superior to S1. Nonetheless, it possessed a comparatively higher deflection of 7.69 mm, reflecting the fact that the composite was capable of bending further before failure. The mid-positioning of paddy straw could have resulted in higher flexibility but less structural stiffness, influencing load-carrying efficiency. Though composed wholly of glass fiber, S4 exhibited flexural strength of 87.393 N/mm², which is almost the same as S1 but unexpectedly low given its fiber type. The deflection of 7.19 mm suggests that the material, though strong, failed to retain structural rigidity as anticipated. This could be attributed to inadequate fiber alignment or bonding during lay-up shown in Figure-16.

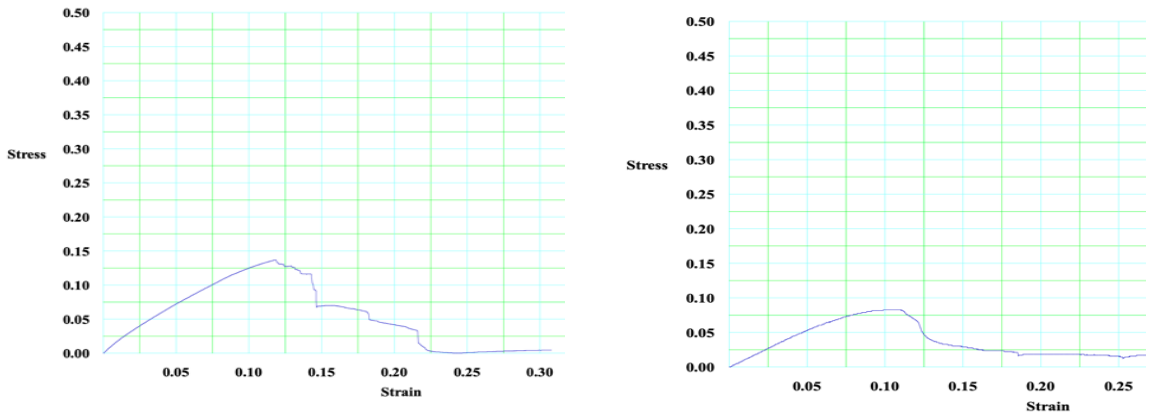


Figure 16 Flexure Stress-Strain plot of sample S3 and S4

This specimen achieved the greatest flexural strength of 123.286 N/mm² and had a moderate deflection of 6.88 mm. Stacking paddy-glass-glass-paddy seems to have established an ideal load path and solid interfacial bonding, hence resulting in superior flexural performance. It is the best overall in the non-Nano series. With a flexural strength of 50.043 N/mm² and deflection of 6.1 mm, S6—made completely of paddy straw—not surprisingly exhibits poorer structural performance. While it provides satisfactory deformability, the lack of any reinforcing fiber such as glass severely restricts its load-carrying ability. It can only be used as a cheaper and environment-friendly option but not for high-load applications shown in Figure-17.

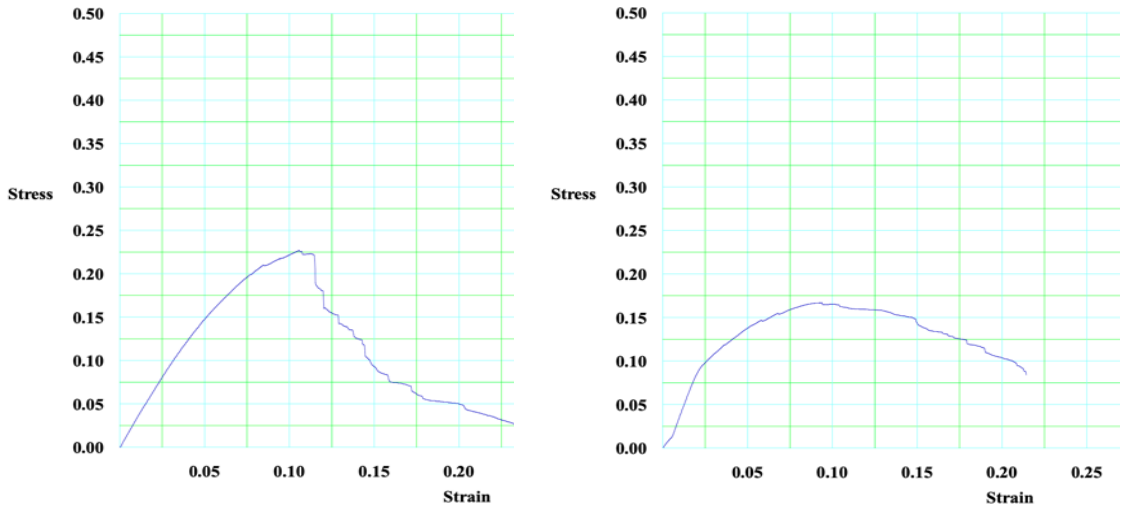


Figure 17 Flexure Stress strain plot of sample S5 and S6

Table 11 Flexure strength of Neat Glass/paddy reinforced epoxy composite (S1-S6)

Specimen no	Stacking Sequence	Thickness (mm)	Peak Load (kg)	Flexture strength (MPa)
S1	GPGPG	2.18±0.15	4	88.456
S2	GPGPG	2.15±0.15	2.96	47.025
S3	GPPG	1.98±0.15	2.98	74.447
S4	GGGG	1.73±0.15	0.63	87.393
S5	PGGP	2.18±0.15	4.36	123.286
S6	PPPP	4.2±0.15	5.83	50.043

Flexture Test without Nano-Silica

Following is a critical comparison of the flexural test results of the Nano-silica reinforced FRP laminates (N1–N6) shown in Table 12 with their respective without Nano silica and the percentage increase in flexural strength:

N1 shows an appreciable improvement in flexural strength with the addition of 2% Nano silica, which would have strengthened the interfacial bonding among the paddy, glass fibers, and epoxy. The deflection reduced slightly (from 4.56 mm to 3.89 mm), indicating a stiffer but stronger composite. This sample demonstrates the successful role of Nano-reinforcement in hybrid laminates. N2 shows a dramatic increase in strength, over two times its flexural strength upon addition of Nano silica. The poor orientation in the non-Nano counterpart (S2) was greatly enhanced by Nano particles, enhancing matrix stiffness and stress transfer. This confirms that even poor stacking can gain significantly from the addition of Nanoparticles shown in Figure-18.

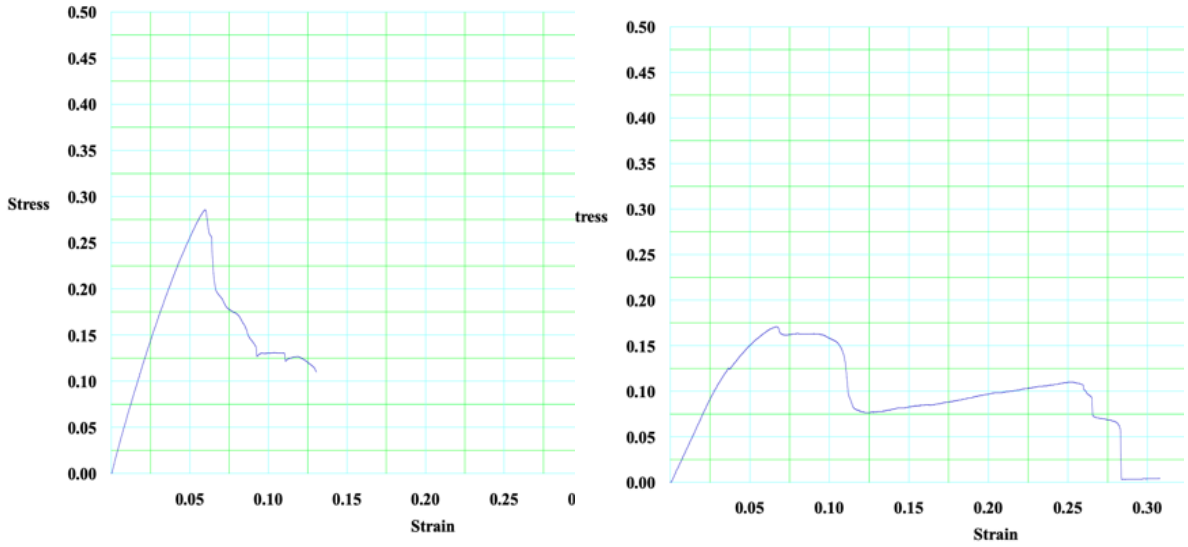


Figure 18 Flexure Stress-Strain plot of sample N1 and N2

Unlike the other samples, N3 unexpectedly indicates a minor reduction in strength even though Nano silica is added. The cause could be poor dispersion or Nanoparticle agglomeration in the matrix, leading to local stress concentrations. Although flexural deflection was enhanced (7.69 mm to 5.04 mm), mechanical strength gain was not achieved. N4 also yielded the highest flexural strength out of all the samples, which meant that Nano silica significantly improved the stiffness as well as the bonding of the all-glass fiber laminate. The vast improvement in strength demonstrates the efficacy with which Nano silica interacts with glass fiber to form a strong, high-performance composite. Deflection also improved, which suggests enhanced toughness. N5 shows an unexpected and drastic drop in strength, indicating that Nano silica may have disrupted the optimal stress transfer in the previously high-performing PGGP stacking. The laminate's performance was likely compromised due to interference with bonding or poor distribution of Nano particles. It highlights that not all stacking sequences benefit from Nano additives. In this 100% paddy-based laminate, addition of Nano silica decreased flexural strength, although deflection increased. The composite became flexible (8.67 mm) but not rigid. Nano particles are not bonded properly with natural fibers such as paddy straw, which may restrict their effectiveness and even decrease strength because of the discontinuities in the matrix shown in Figure-19 & Figure-20.

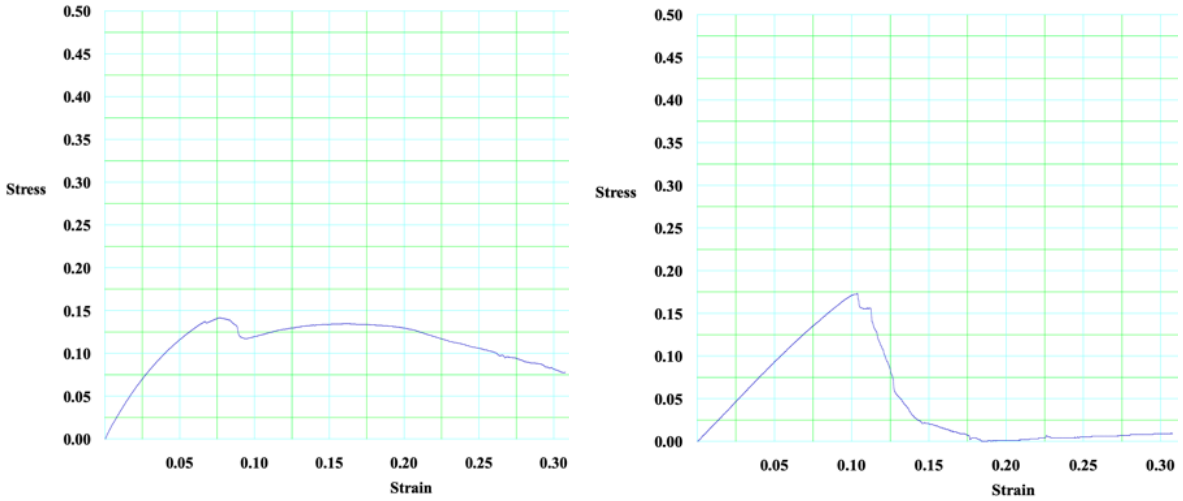
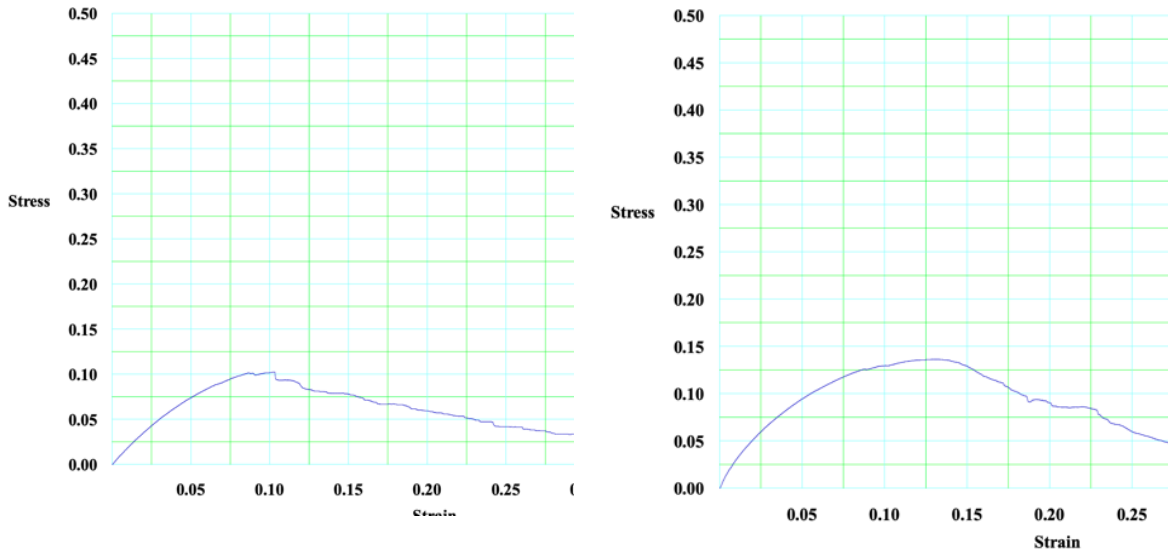


Figure 19 Flexure Stress-Strain plot of sample N3 and N4



Stress-Strain curve of sample N5

Stress-Strain curve of sample N6

Figure 20 Flexure Stress-Strain of sample N5 and N6

Table 12 Flexure strength of modified Glass/paddy reinforced epoxy composite with Nano silica (N1-N6)

Specimen no	Stacking Sequence	Thickness mm	Peak Load kg	Flexture strength (MPa)
N1	GPGPG	2.11±0.15	7.75	121.892
N2	GPGPG	1.85±0.15	2.97	97.847
N3	GPPG	2.08±0.15	2.76	71.542
N4	GGGG	1.07±0.15	1.73	159.361
N5	PGGP	2.06±0.15	2.38	40.573
N6	PPPP	3.06±0.15	4.78	43.56

Water absorption test

Following is a critical discussion of the water absorption test results of the FRP laminates without and with 2% addition of Nano silica based on 24-hour immersion:

Water Absorption Behavior – Non-Nano Samples (S1–S6):

The non-Nano FRP laminates have a variety of water absorption percentages, reflecting differing resistance against moisture intrusion based on the sequence of stacking and fiber content.

•S1 (GPGPG) and S4 (GGGG) exhibited 0% water absorption, which is indicative of high water resistance. The use of glass fibers in more than one layer significantly improves impermeability owing to their low porosity and good bonding with the resin matrix.

•S2 (GPGPG), which has the same stack sequence as S1, absorbed 9.09%, indicating probable voids or fabrication inaccuracies in resin impregnation or quality of lamination.

•S3 (GPPG) absorbed 16.6%, and S5 (PGGP) absorbed 25%, revealing that increasing the presence of paddy fibers, which are hydrophilic, led to higher water uptake.

•S6 (PPPP) had the highest absorption at 27.27%, which is expected as it is composed entirely of paddy fibers, which readily absorb moisture due to their natural structure and lower bonding with epoxy.

Water Absorption Behavior – Nano Silica-Reinforced Samples (N1–N6):

Addition of 2% Nano silica demonstrates mixed behavior, enhancing moisture resistance in certain samples while elevating absorption in others based on stacking design.

•N4 (GGGG) stayed at 0% absorption, the same as S4, and affirmed that Nano silica does not have a negative effect on the high resistance of full-glass laminates.

•N1 (GPGPG) registered a high increase to 20% from 0% (S1), which is anomalous. The likely reason could be Nanoparticle agglomeration or enhanced surface area in the matrix, inadvertently forming micro voids.

•N2 (GPGPG) has incomplete data, so assessment remains incomplete; however, the earlier result indicates variability in stacking for the same sequences.

•N3 (GPPG) experienced 10% absorption, a gain over 16.6% in S3, reflecting positive influences of Nano silica in decreasing porosity and enhancing the matrix-fiber interface for this sequence.

•N5 (PGGP) remained stable at 25%, the same as S5, reflecting no significant influence of Nano silica on water absorption for this configuration.

•N6 (PPPP) increased to 33.3%, from 27.27%, suggesting that Nano silica could worsen water absorption in paddy-based laminates because of inadequate bonding and higher surface activity.

- Glass fiber-dominant laminates (S1, S4, N4) show excellent moisture resistance, regardless of Nano silica addition.

- Nano silica improved water resistance in some hybrids (e.g., N3), but increased absorption in others (N1, N6), suggesting that Nano dispersion quality and matrix compatibility are crucial.
- Fully natural-fiber laminates (S6, N6) remain highly susceptible to moisture and even worsen with Nano addition, likely due to poor compatibility between Nano silica and natural fibers.
- The inconsistencies in same stacking sequences (S1 vs. S2, N1 vs. N2) highlight the importance of manufacturing precision and potential issues like resin-rich/poor zones or voids.

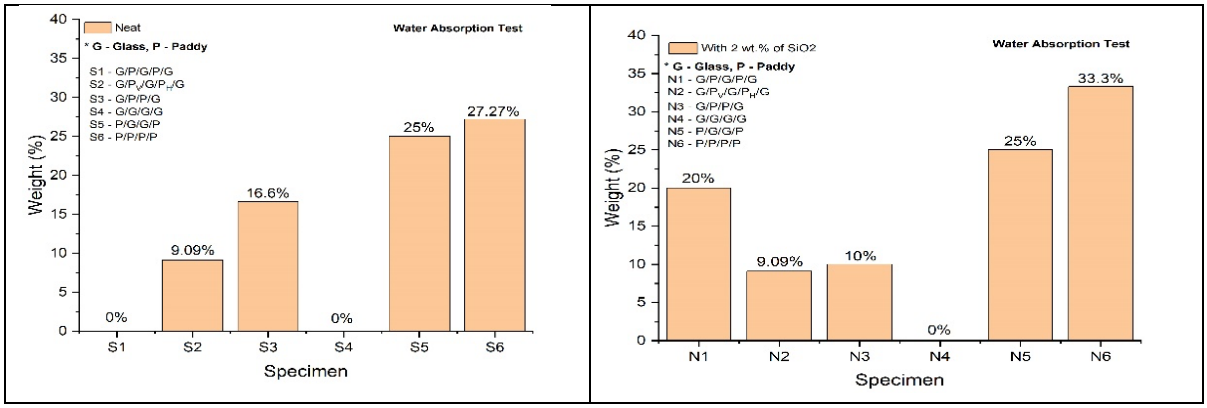


Figure 21 Water absorption test results of sample without Nano silica (S1-S6) & with Nano silica (N1-N6)

CONCLUSION

Tensile, flexural, and water absorption tests were conducted on both neat glass/paddy reinforced epoxy composites and modified composites with 2 wt.% nano-silica. The results demonstrated significant improvements in mechanical and physical performance with specific stacking sequences and nano-silica modification. Among all configurations, S4 (G/G/G/G) exhibited the highest tensile strength with an impressive 198% increase compared to the modified hybrid composite, owing to the superior load-bearing capacity and uniform stress distribution of the all-glass fiber structure. The S3 configuration (G/P/P/G) also showed notable enhancement, recording a 104% increase in tensile strength, indicating the positive influence of symmetrical stacking and glass outer layers.

In flexural testing, the S2 configuration (G/Pv/G/Ph/G) achieved the best performance with a 108% increase in flexural strength. This can be attributed to the mixed orientation of paddy fibers (vertical and horizontal), which effectively resisted bending forces and enhanced interlaminar bonding with the glass layers.

Water absorption behavior revealed that the S5 configuration (P/G/G/P) reduced water uptake by 10%, while the S4 configuration (G/G/G/G) exhibited zero water absorption, highlighting the excellent moisture resistance of glass fiber and the protective effect of its continuous outer layers.

Overall, the incorporation of Nano-silica and strategic fiber stacking significantly improved the composite's mechanical strength and water resistance, making the S4 configuration the most promising for structural applications requiring high durability and environmental resistance.

REFERENCE

1. Gupta, M. Singh, and J. S. Saini, "Effect of Nano-silica on fatigue behavior of glass fiber-reinforced epoxy composite laminates: A Weibull distribution approach," Polymer Composites, vol. 44, no. 12, pp. 9129–9140, Dec. 2023.
2. M. J. Ghasemi Parizi et al., "Improving Mechanical Properties of Glass Fiber Reinforced Polymers through Silica-Based Surface Nanoengineering," ACS Applied Polymer Materials, vol. 2, no. 7, pp. 2667–2675, Jul. 2020.
3. M. G. M. Ali, M. A. M. Ali, and M. A. M. Ali, "Mechanical Properties and Water Absorption Capacity of Hybrid GFRP Composites," Polymers, vol. 14, no. 7, p. 1394, Mar. 2022.
4. G. Ganesan, S. Muthunadar, and M. Ramasamy, "Experimental investigation on mechanical properties and free vibration characteristics of epoxy-based glass/flax laminates," Polymer Composites, vol. 43, no. 9, pp. 6004–6012, Sep. 2022.
5. R. Verma et al., "Experimental assessment of Nanoclay effect on hardness, tensile and flexural properties of glass fiber/epoxy laminates," Materials Today: Proceedings, vol. 79, pp. 395–397, 2023.
6. M. Vijayan et al., "Low-Velocity Impact Response of Nano-Silica Reinforced Aluminum/PU/GFRP Laminates," in Dynamic Behavior of Soft and Hard Materials Volume 1, R. Velmurugan, G. Balaganesan, N. Kakur, and K. Kanny, Eds. Singapore: Springer, 2024, pp. 467–478.
7. Su, X. Wang, and Z. Wu, "Enhancement of Mechanical Properties of FRP Composites with Silica Nanoparticles," in 10th International Conference on FRP Composites in Civil Engineering, Istanbul, Turkey, 2021, pp. 684–694.
8. M. S. R. Theja et al., "Investigation into mechanical properties of EPDM/SBR-Nanoclay Nanocomposites," Materials Today: Proceedings, vol. 59, pp. 1508–1512, 2022.
9. R. Verma et al., "Experimental assessment of Nanoclay effect on hardness, tensile and flexural properties of glass fiber/epoxy laminates," Materials Today: Proceedings, vol. 79, pp. 395–397, 2023.

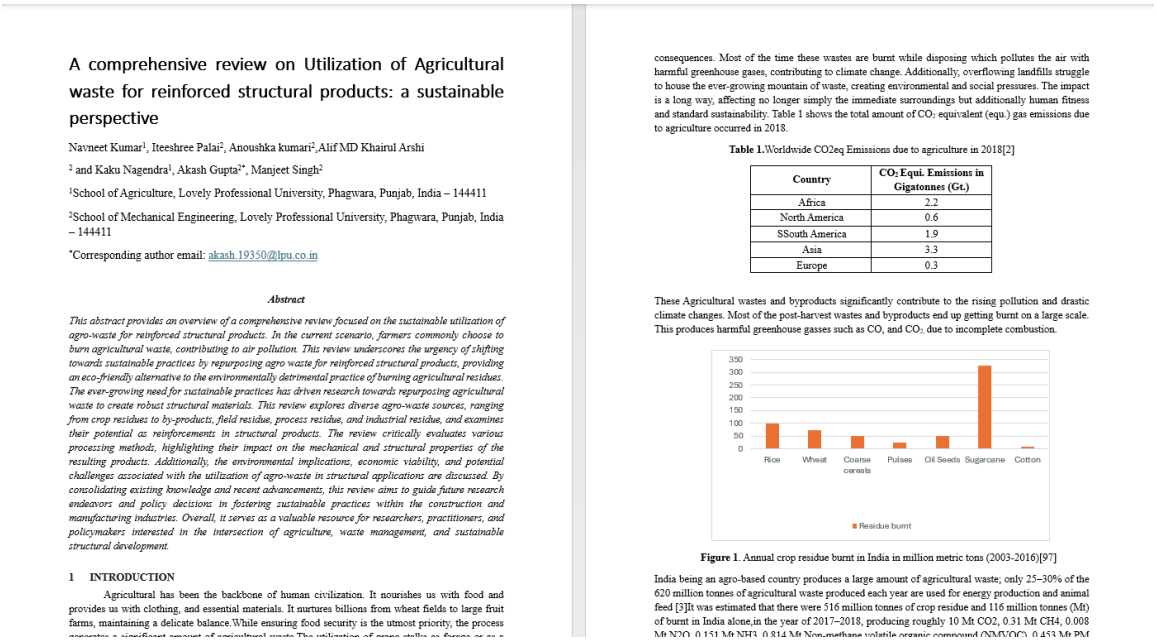
10. M. J. Ghasemi Parizi et al., "Improving Mechanical Properties of Glass Fiber Reinforced Polymers through Silica-Based Surface Nanoengineering," ACS Applied Polymer Materials, vol. 2, no. 7, pp. 2667–2675, Jul. 2020.
11. Gupta, M. Singh, and J. S. Saini, "Effect of Nano-silica on fatigue behavior of glass fiber-reinforced epoxy composite laminates: A Weibull distribution approach," Polymer Composites, vol. 44, no. 12, pp. 9129–9140, Dec. 2023.
12. G. Ganesan, S. Muthunadar, and M. Ramasamy, "Experimental investigation on mechanical properties and free vibration characteristics of epoxy-based glass/flax laminates," Polymer Composites, vol. 43, no. 9, pp. 6004–6012, Sep. 2022.
13. M. G. M. Ali, M. A. M. Ali, and M. A. M. Ali, "Mechanical Properties and Water Absorption Capacity of Hybrid GFRP Composites," Polymers, vol. 14, no. 7, p. 1394, Mar. 2022.
14. Su, X. Wang, and Z. Wu, "Enhancement of Mechanical Properties of FRP Composites with Silica Nanoparticles," in 10th International Conference on FRP Composites in Civil Engineering, Istanbul, Turkey, 2021, pp. 684–694.
15. M. Vijayan et al., "Low-Velocity Impact Response of Nano-Silica Reinforced Aluminum/PU/GFRP Laminates," in Dynamic Behavior of Soft and Hard Materials Volume 1, R. Velmurugan, G. Balaganesan, N. Kakur, and K. Kanny, Eds. Singapore: Springer, 2024, pp. 467–478.
16. Maraveas, C. (2024), "A review on the use of agro-waste in sustainable building materials," ScienceDirect, 2012.
17. K. Vinod, S. Das, and K. Prabhakaran, "Tribological behavior of hybrid aluminum matrix composites reinforced with agro-industrial waste," SpringerLink, 2019.
18. V. Dwivedi and S. Mishra, "Physico-chemical, mechanical, and thermal properties of RHA green composites," SpringerLink, 2019.
19. O. P. Durowaye, B. Adewuyi, and J. O. Akintayo, "Mechanical properties of hybrid unsaturated polyester matrix composites reinforced with agro-waste particles," Nepal Journals Online, 2019.

20. M. P. Soh, H. W. Tan, and C. K. Ho, "Extrudable mycelium-bound composites from agricultural waste," arXiv preprint, 2020.
21. P. Raut, R. Sedmake, and J. Mandavgane, "Utilization of agro-waste in building materials: A review," ScienceDirect, 2012.
22. T. Kılıç, "Environmentally friendly brake friction composites based on rice husk and rice stalk wastes," 2024.
23. Kellersztein, L. Zhegur-Karady, and S. Kenig, "Improving properties of wood plastic composites using treated wheat straw fibers," SpringerLink, 2019.
24. J. Du, H. Zhao, and Y. Wang, "Green filler from peanut shells for polyamide biocomposites," 2022.
25. Laad, "Reinforcement of aluminum matrix composites with rice husk ash through stir casting method," 2019.

APPENDIX

Review paper (communicated in Bulgarian Chemical Communications Journal)

A comprehensive review on Utilization of Agricultural waste for reinforced structural products: a sustainable perspective



A comprehensive review on Utilization of Agricultural waste for reinforced structural products: a sustainable perspective

Navneet Kumar[1], Iteeshree Palai[2], Anoushka kumari[2],Alif MD Khairul Arshi

[2] and Kaku Nagendra[1], Akash Gupta[2]*, Manjeet Singh[2]

[1]School of Agriculture, Lovely Professional University, Phagwara, Punjab, India – 144411

[2]School of Mechanical Engineering, Lovely Professional University, Phagwara, Punjab, India – 144411

*Corresponding author email: akash.19350@lpu.co.in

Abstract

This abstract provides an overview of a comprehensive review focused on the sustainable utilization of agro-waste for reinforced structural products. In the current scenario, farmers commonly choose to burn agricultural waste, contributing to air pollution. This review underscores the urgency of shifting towards sustainable practices by repurposing agro waste for reinforced structural products, providing an eco-friendly alternative to the environmentally detrimental practice of burning agricultural residues. The ever-growing need for sustainable practices has driven research towards repurposing agricultural waste to create robust structural materials. This review explores diverse agro-waste sources, ranging from crop residues to by-products, field residue, process residue, and industrial residue, and examines their potential as reinforcements in structural products. The review critically evaluates various processing methods, highlighting their impact on the mechanical and structural properties of the resulting products. Additionally, the environmental implications, economic viability, and potential challenges associated with the utilization of agro-waste in structural applications are discussed. By consolidating existing knowledge and recent advancements, this review aims to guide future research endeavors and policy decisions in fostering sustainable practices within the construction and manufacturing industries. Overall, it serves as a valuable resource for researchers, practitioners, and policymakers interested in the intersection of agriculture, waste management, and sustainable structural development.

1 INTRODUCTION

Agricultural has been the backbone of human civilization. It nourishes us with food and provides us with clothing, and essential materials. It nurtures billions from wheat fields to large fruit farms, maintaining a delicate balance.While ensuring food security is the utmost priority, the process

consequences. Most of the time these wastes are burnt while disposing which pollutes the air with harmful greenhouse gases, contributing to climate change. Additionally, overflowing landfills struggle to house the ever-growing mountain of waste, creating environmental and social pressures. The impact is a long way, affecting no longer simply the immediate surroundings but additionally human fitness and standard sustainability. Table 1 shows the total amount of CO_2 equivalent (equ.) gas emissions due to agriculture occurred in 2018.

Table 1. Worldwide CO2eq Emissions due to agriculture in 2018[2]

Country	CO_2 Equi. Emissions in Gigatonnes (Gt.)
Africa	2.2
North America	0.6
SSouth America	1.9
Asia	3.3
Europe	0.3

These Agricultural wastes and byproducts significantly contribute to the rising pollution and drastic climate changes. Most of the post-harvest wastes and byproducts end up getting burnt on a large scale. This produces harmful greenhouse gasses such as CO, and CO_2 due to incomplete combustion.

Figure 1. Annual crop residue burnt in India in million metric tons (2003-2016)[97]

India being an agro-based country produces a large amount of agricultural waste; only 25–30% of the 620 million tonnes of agricultural waste produced each year are used for energy production and animal feed [3]It was estimated that there were 516 million tonnes of crop residue and 116 million tonnes (Mt) of burnt in India alone,in the year of 2017–2018, producing roughly 10 Mt CO2, 0.31 Mt CH4, 0.008 Mt N2O, 0.151 Mt NH3, 0.814 Mt Non-methane volatile organic compound (NMVOC), 0.453 Mt PM

Abstract

This abstract provides an overview of a comprehensive review focused on the sustainable utilization of agro-waste for reinforced structural products. In the current scenario, farmers commonly choose to burn agricultural waste, contributing to air pollution. This review underscores the urgency of shifting towards sustainable practices by repurposing agro waste for reinforced structural products, providing an eco-friendly alternative to the environmentally detrimental practice of burning agricultural residues. The ever-growing need for sustainable practices has driven research towards repurposing agricultural waste to create robust structural materials. This review explores diverse agro-waste sources, ranging from crop residues to by-products, field residue, process residue, and industrial residue, and examines their potential as reinforcements in structural products. The review critically evaluates various processing

methods, highlighting their impact on the mechanical and structural properties of the resulting products. Additionally, the environmental implications, economic viability, and potential challenges associated with the utilization of agro-waste in structural applications are discussed. By consolidating existing knowledge and recent advancements, this review aims to guide future research endeavors and policy decisions in fostering sustainable practices within the construction and manufacturing industries. Overall, it serves as a valuable resource

Patent under filing

Annexure (Provisional Patent)

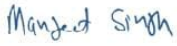
INVENTION DISCLOSURE FORM

Details of Invention for better understanding:

1. TITLE: Sustainable, Lightweight, and Durable Agro-Waste Reinforced Glass/Epoxy Composite for False Ceiling Applications.

2. INTERNAL INVENTOR(S)/ STUDENT(S): All fields in this column are mandatory to be filled

A.	Full name	Anoushka Kumari
	Mobile Number	9162152352
	Email (personal)	kumarianoushka2@gmai.com
	UID/Registration number	12115465
	Address of Internal Inventors	Lovely Professional University, Punjab-144411, India
	Signature (Mandatory)	
B.	Full name	Iteeshree Palai
	Mobile Number	7008102922
	Email (personal)	iteeshreepalai@gmail.com
	UID/Registration number	12110765
	Address of Internal Inventors	Lovely Professional University, Punjab-144411, India
	Signature (Mandatory)	
C.	Full name	Alif MD Khairul Arshi
	Mobile Number	8360331967
	Email (personal)	arshialif001@gmail.com

UID/Registration number	12110776
Address of Internal Inventors	Lovely Professional University, Punjab-144411, India
Signature (Mandatory)	
D. Full name	Dr. Akash Gupta
Mobile Number	9501994809
Email (personal)	akash2588@gmail.com
UID/Registration number	19350
Address of Internal Inventors	Lovely Professional University, Punjab-144411, India
Signature (Mandatory)	
E. Full name	Dr. Manjeet Singh
Mobile Number	8198800563
Email (personal)	manjeetsekhon21545@lgmail.com
UID/Registration number	21545
Address of Internal Inventors	Lovely Professional University, Punjab-144411, India
Signature (Mandatory)	
F. Full name	Dr. Manish Dhawan
Mobile Number	7814433339
Email (personal)	mds_78@rediffmail.com
UID/Registration number	14630